O MIRROR, WHERE ART THOU?

by

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DEDICATION

Dedicated to

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ABSTRACT

Wavefront sensing reveals that a telescope is misaligned, but not which mirror moved or by how much; inferring the mirror positions themselves turns alignment into a direct correction. We describe a simulation framework for inferring telescope mirror misalignment from multi-field Point Spread Function (PSF) measurements projected onto Circular-Bessel and Zernike basis vectors. Two case studies are presented: a Hubble Space Telescope model with five secondary-mirror degrees of freedom and a three-mirror anastigmat with ten primary and secondary mirror degrees of freedom. In both cases, field-dependent multivariate Gaussian likelihoods are explored with nested sampling and refined with differential evolution. Recovered states are validated with residual PSF images and optical path difference (OPD) measurements. In the Hubble benchmark with noisy PSFs, differential evolution recovers the injected translations to $\lesssim 4\,\mu$m and reduces on-axis OPD error from ~ 2.3 waves to $\sim 1.3 \times 10^{-3}$ waves. In the TMA study, the posterior shows broad marginals and correlated parameter pairs that suggest partial degeneracies at higher dimension; differential evolution recovers the truth alignment state less precisely than in the Hubble benchmark (translation residuals up to $\sim 46\,\mu$m) and reduces on-axis OPD error from ~ 8.24 waves to $\sim 2.5 \times 10^{-3}$ waves. A linearized noise analysis, checked against Monte Carlo simulations, shows that in both benchmarks the residuals are those expected from the observation noise, amplified along near-degenerate combinations of mirror motions, and that the errors grow in proportion to the noise up to 3×10^{-3} of the ideal PSF peak for Hubble and 3×10^{-4} for the TMA. Observing the Hubble secondary again after known tilts of ± 0.1 degrees makes the coefficients more sensitive to the alignment: it reduces the noise-limited errors by a further factor of about three beyond the gain from the additional exposures, and a full run recovers both decenters to within $0.2\,\mu$m. This generalizable algorithm enables direct sensing of the state of an optical system and automated alignment.

Chapter 1

Introduction

Optical observatories depend on precise alignment of telescope optics to maintain image quality. Wavefront aberrations from mirror translations and tilts propagate into field-dependent point spread functions (PSFs). These field-dependent PSFs therefore provide observables sensitive to rigid-body misalignment even when direct wavefront sensing is unavailable or incomplete.

How a misalignment appears in the image depends on the optical design. In a rotationally symmetric telescope the aberration field contributed by each surface is centered on the optical axis. Decentering or tilting a mirror displaces that surface's contribution, and the combined aberration field develops nodes, field points of zero aberration, away from the field center. Nodal aberration theory (Thompson, 2005) describes this behavior at third order, and Thompson et al. (2008) applied it to the misalignment-induced aberrations of three-mirror anastigmats. Schechter and Levinson (2011) showed that the resulting patterns are generic: each is set by a two-dimensional vector built from the decenters and tilts of the individual mirrors, and keeping an N-mirror telescope aligned requires measuring $2(N-1)$ of these patterns. They also found that a three-mirror anastigmat has a subspace of misalignments that introduces no third-order coma, astigmatism or field curvature. Different combinations of mirror motions can therefore produce nearly the same image, which is one reason a single field point cannot identify a misalignment and why the likelihood in this work combines several field points.

In this work, we implement a simulation-based solution to the inverse problem of estimating mirror positions from observables: given multi-field PSF observations projected onto basis vectors, we estimate the underlying telescope mirror alignment parameters $\boldsymbol{\theta}$.

Table 1.1: Summary of the two telescope models used in this study.

Telescope	Degrees of freedom	Noise	Basis (modes)	Fields	Wavelength
Hubble	5	Yes	Circular-Bessel (15)	5	0.55 μm
TMA	10	Yes	Zernike (11)	10	0.65 μm

Synthetic PSFs generated from an open-source ray-tracing model are compared to stored "observations" through a multivariate Gaussian likelihood in coefficient space. We present benchmarks against example two and three mirror telescope models as summarized in Table 1.1.

Historical HST servicing relied on phase retrieval with focus diversity (Fienup, 1982; Löfgren et al., 2014; Krist and Burrows, 1997) rather than direct PSF-to-alignment inversion leveraging a full ray-trace model. The commissioning of the James Webb Space Telescope (JWST), a deployable three-mirror anastigmat (Knight et al., 2012), is the most relevant recent precedent. Because the segmented primary can compensate the on-axis wavefront error caused by a misplaced secondary mirror, secondary-mirror misalignment appears mainly in field-dependent terms (astigmatism and focal-plane tilt) that can only be sensed at multiple field points (Acton and Knight, 2012). The final stage of JWST telescope commissioning, Multi-Instrument Multi-Field (MIMF) alignment, measured the wavefront with phase retrieval at field points across the science instruments to verify and refine the secondary-mirror position (Acton et al., 2022; McElwain et al., 2023). That approach reconstructs the wavefront at each field point and maps it to mirror corrections through a linear optical model; here we instead infer rigid-body mirror positions directly from PSF intensity with a non-linear ray-trace forward model, obtaining a posterior over the alignment parameters.

Ground-based observatories and laboratory systems use related approaches. A common approach converts measured Zernike coefficients into mirror corrections through a linear sensitivity matrix, or sensitivity table, computed from the optical design. Merit-function regression (Kim et al., 2007) instead fits the misalignment parameters by damped least squares, which copes better with non-linear wavefront sensitivity. Phase diversity (Gonsalves, 1982) estimates the wavefront from images taken with a known added aberration, typically defocus. The Rubin Observatory measures its wavefront with curvature sensors at the corners of the focal plane and feeds the estimates to an active optics system that corrects the align-

ment and figure of its optics (Xin et al., 2015). Each of these methods first estimates the wavefront and then infers the alignment through a model of how the wavefront responds to mirror motions, usually linearized about the nominal design.

Modern open-source ray tracers and nested samplers make it practical to evaluate a full forward model at every likelihood call over a large enough parameter space for the method to be practically useful. Missions such as the Habitable Worlds Observatory (HWO; NASA Goddard Space Flight Center 2023) and the Lazuli Space Observatory (Roy et al., 2026) will require precise and robust alignment algorithms for complex three-mirror anastigmat designs. We investigate the feasibility of solving for telescope misalignments directly from PSF observables as a method for future observatories, like HWO, to rapidly and precisely stabilize their mirrors.

This thesis makes the following contributions:

1. A forward model that ray traces rigid-body mirror misalignments in Optiland and reduces each PSF to a short vector of basis coefficients, so that a full non-linear model can be evaluated at every likelihood call.

2. A two-stage inference pipeline in which nested sampling explores the posterior over the alignment parameters and differential evolution refines a point estimate within bounds taken from that posterior.

3. Benchmarks on a two-mirror Hubble model with five secondary-mirror degrees of freedom and on a three-mirror anastigmat with ten primary- and secondary-mirror degrees of freedom, validated by comparing PSFs and OPD maps with the nominal designs.

4. An analysis of the parameter degeneracies in the TMA posterior and of their optical origin.

5. An analysis of how observation noise propagates through the basis coefficients into the likelihood and the differential-evolution estimate. It shows that the residuals of both benchmarks are those predicted by the noise, amplified along the near-degenerate directions, and how the errors scale with the noise level.

6. A study, with the linearized analysis and with full runs of the pipeline, of observations taken after known moves of the Hubble secondary. Tilts of ± 0.1 degrees reduce the noise-limited errors by a further factor of about 3, and 7 on the despace, beyond the gain from the additional exposures, by making the coefficients more sensitive to the alignment rather than by breaking the decenter–tilt degeneracy.

The rest of the thesis is organized as follows. Chapter 2 describes the forward model, the likelihood, nested sampling and differential evolution. Chapters 3 and 4 apply the pipeline to the Hubble and TMA models and present the results. Chapter 5 examines how noise affects both models and whether known moves of the Hubble secondary improve the precision, and Chapter 6 summarizes the results and outlines future work.

Portions of Chapters 1–4 and 6 are adapted from a co-authored manuscript, "O Mirror, Where Art Thou? Position retrieval using nested sampling and global minimization", by the author with S. Sabhlok, J. N. Ashcraft, K. Van Gorkom and E. S. Douglas, prepared for RAS Techniques and Instruments.

Chapter 2

Methods

Misalignments produce aberrations, but phase retrieval recovers only the wavefront; inferring the positions of the telescope optics requires a second, model-based step, such as the linear multi-field inversion used during JWST commissioning (Chapter 1). In this work, we instead use a ray-trace model of the telescope to infer mirror misalignments directly from PSF observations.

To perform misalignment inference, we construct a pipeline (Figure 2.1) that does the following:

1. Build a parametric telescope in Optiland (Harrison, 2026) and propagate monochromatic light to generate a synthetic PSF.
2. Project PSF onto a basis.
3. Model observations as multivariate Gaussians in coefficient space with field-specific covariances $\mathbf{\Sigma}_f$ from Monte Carlo calibration.
4. Explore the posterior with Nautilus (Lange, 2023), and then refine a point estimate with differential evolution.
5. Validate by comparing nominal and residual-alignment systems in PSF and OPD space.

2.1 Raytracing

Both telescope models are built and traced in Optiland (Harrison, 2026) (version 0.6.0; see Section 2.4), an open-source Python ray tracer, using its NumPy backend in double

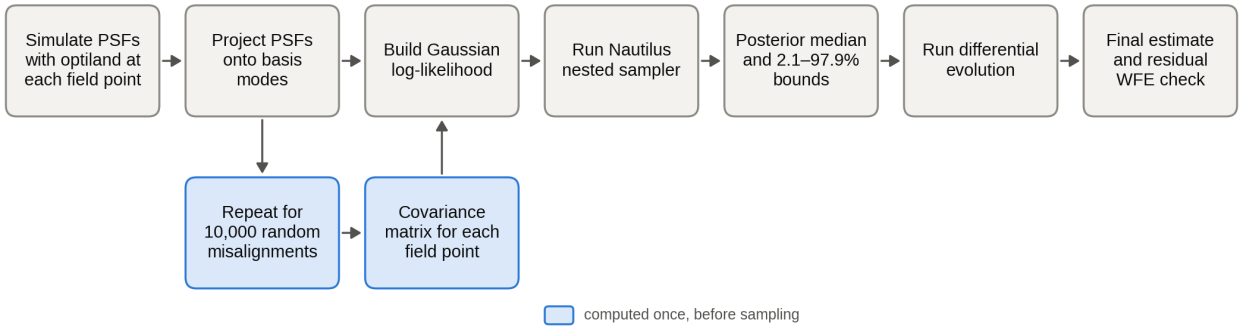


Figure 2.1: Inference pipeline: Optiland forward model, basis-coefficient feature extraction, multivariate Gaussian likelihood, Nautilus nested sampling, and differential-evolution refinement with PSF/OPD validation.

Table 2.1: Optical prescriptions of the two telescope models. Vertex positions z are along the optical axis in the fixed laboratory frame used to place each surface.

Model	Surface	R (mm)	k	z (mm)
Hubble	Primary (stop)	-11040.02286	-1.001152	4910.01016
	Secondary	-1349.31166	-1.483014	0
	Image	-635.38227	—	6365.20955
TMA	M1 (stop)	-16256	-0.995357	0
	M2	-1660.575	-1.566503	-7400.005
	M3	-1830.5174	-0.718069	49.965
	M4 (fold)	∞	—	-983.655
	Image	∞	—	236.685

precision. Each mirror is a conic surface with sag

$$z(r) = \frac{r^2/R}{1 + \sqrt{1 - (1+k)\, r^2/R^2}}, \tag{2.1}$$

where R is the vertex radius of curvature, k is the conic constant and r is the radial distance from the vertex. Real rays are traced sequentially from an object at infinity and intersect each surface exactly rather than through a paraxial approximation, so the forward model contains every order of aberration that a misalignment introduces. Lengths are in millimeters and wavelengths in micrometers. Table 2.1 lists both prescriptions.

2.1.1 Telescope models

The Hubble model is a Ritchey–Chrétien telescope with a 2400 mm entrance pupil. The primary is the aperture stop and carries a physical mask: an annulus between radii of 384 and

1200 mm, two orthogonal spider vanes 38 mm wide, and three circular pads of 75 mm radius centered 1000 mm from the axis. The secondary sits 4910.01 mm in front of the primary and the curved image surface lies 6365.21 mm behind the secondary, giving a working f-number of 24.0. Field angles extend to 0.038°.

The TMA follows the design of Kim et al. (2023). Three powered conic mirrors form the image, and a flat fourth mirror folds the beam onto the detector, which is why the Optiland prescription has four mirrors. The entrance pupil is circular with a diameter of 6471.26 mm, the primary is the stop, and the working f-number is 15.1. The prescription defines field angles of 0.055°, 0.17° and 0.285° along y. Normalized field coordinates are relative to 0.285°, so the near-axis field $(0, 0.5965)$ used in the likelihood corresponds to 0.17°.

2.1.2 Applying misalignments

Rather than chaining surfaces through sequential thicknesses, every surface is placed in absolute coordinates: its vertex position (x, y, z) and rotations (r_x, r_y) are given in a fixed laboratory frame. A rigid-body misalignment of a mirror then changes only that mirror's coordinates. Decenters add to its x and y, despace adds to its z, and tilts rotate it about its own vertex. All other surfaces, including the image surface, stay where they are, so moving the secondary does not drag the downstream optics or the detector with it, as it would if a despace were entered as a change in thickness. Misalignments are specified in micrometers and degrees and converted to millimeters and radians before tracing. The Hubble model perturbs only the secondary (*sdx*, *sdy*, *sdz*, *stx*, *sty*). The TMA perturbs the primary and the secondary (*pdx*, *pdy*, *pdz*, *ptx*, *pty* and *sdx*, *sdy*, *sdz*, *stx*, *sty*), while M3, the fold mirror and the detector stay fixed.

2.1.3 PSF computation

PSFs are computed with Optiland's `MMDFTPSF`. For a given field and wavelength it traces a uniform square grid of rays across the entrance pupil and computes the optical path difference W of each ray, in waves, at the exit pupil relative to a reference sphere centered on the chief ray's image point. Each PSF is therefore centered on the chief ray: a misalignment that moves the whole image, and the chief ray with it, does not move the PSF within the frame.

Table 2.2: PSF sampling for the two models.

	Hubble	TMA
Wavelength	$0.55\,\mu$m	$0.65\,\mu$m
Working f-number $F_\#$	24.0	15.1
`num_rays` N	128	512
Pupil samples	64×64	128×128
Image grid	256×256	1024×1024
Pixel pitch	$3.25\,\mu$m	$1.21\,\mu$m
Pixels per $\lambda F_\#$	4.06	8.06
Retained patch	64×64 ($15.8\,\lambda F_\#$)	256×256 ($31.8\,\lambda F_\#$)

Tilt relative to the reference sphere is retained, so displacements of the light relative to the chief ray, such as the centroid shift produced by coma, do appear. The absolute position of the image on the detector is not used (Section 5.5). The complex pupil function is

$$P(x, y) = A(x, y)\, \exp[-2\pi i\, W(x, y)]\,, \tag{2.2}$$

where the amplitude A follows from the transmitted ray intensity and is zero for rays blocked by the aperture mask. The focal-plane field is computed with a matrix-triple-product discrete Fourier transform (Soummer et al., 2007), $E = L\,P\,R$, in which the matrices L and R set the image sampling independently of the pupil grid, and the PSF is $|E|^2$. Optiland scales the PSF so that a diffraction-limited system with the same aperture has a peak of 100. The peak of each PSF is therefore 100 times its Strehl ratio, and the noise level $\sigma = 0.01$ used for the observations of both models (Sections 3.1 and 4.1) is 0.01% of the ideal peak.

The `num_rays` argument N sets both samplings, following the OpticStudio convention. The pupil is sampled with $n_\mathrm{p} = \lfloor 32 \cdot 2^{(\log_2 N - 5)/2} \rfloor$ rays across its diameter, the image grid is $2N \times 2N$ pixels, and the pixel pitch is chosen so that the grid spans $(n_\mathrm{p} - 1)\,\lambda F_\#$. Only a central patch of the image is kept for the basis projection. Table 2.2 lists the resulting values for both models. The PSFs are monochromatic and contain no detector effects such as pixel integration, jitter or read noise.

2.1.4 Wavefront error

Recovered alignments are validated by comparing optical path difference maps. Optiland's `OPD` traces a hexapolar ray pattern of 15 rings at the requested field, computes the OPD in waves relative to the same chief-ray reference sphere, and interpolates it onto a 256×256 pupil grid. The RMS wavefront error quoted in Chapters 3 and 4 is the standard deviation, over the pupil, of the difference between the OPD map of the nominal design and that of the residual system with misalignments $\boldsymbol{\theta}_{\text{true}} - \mathbf{c}$. Taking the standard deviation removes piston; tilt is retained.

2.2 Nested sampling

2.2.1 Bayesian inference

We treat the misalignment $\boldsymbol{\theta}$ as uncertain and describe what the observations imply about it with Bayes' theorem,

$$P(\boldsymbol{\theta} \mid D) = \frac{\mathcal{L}(\boldsymbol{\theta})\,\pi(\boldsymbol{\theta})}{\mathcal{Z}}, \qquad \mathcal{Z} = \int \mathcal{L}(\boldsymbol{\theta})\,\pi(\boldsymbol{\theta})\,\mathrm{d}\boldsymbol{\theta}, \tag{2.3}$$

where D denotes the observed coefficients, $\mathcal{L}(\boldsymbol{\theta}) = P(D \mid \boldsymbol{\theta})$ is the likelihood, $\pi(\boldsymbol{\theta})$ is the prior, and the evidence $\mathcal{Z}$ normalizes the posterior. The posterior says more than a single best fit. Its marginals show how well each parameter is constrained, and correlations between parameters reveal combinations of mirror motions that produce nearly the same PSFs (Chapters 3 and 4). The evidence is the quantity used to compare models, for example telescope models with different sets of free parameters; we report it for each run but do not use it for model selection here.

With five or ten parameters, the posterior cannot be mapped on a grid. A misaligned telescope can also produce a likelihood surface with separated maxima or long curved ridges, which are difficult for local optimizers and slow the mixing of Markov chain Monte Carlo samplers. Nested sampling (Skilling, 2006) was designed to compute $\mathcal{Z}$ for such problems, and it yields posterior samples as a by-product.

2.2.2 The nested sampling algorithm

Nested sampling turns the multidimensional evidence integral into a one-dimensional one. The prior volume enclosed by the likelihood contour λ,

$$X(\lambda) = \int_{\mathcal{L}(\boldsymbol{\theta})>\lambda} \pi(\boldsymbol{\theta})\,\mathrm{d}\boldsymbol{\theta}, \tag{2.4}$$

falls from 1 to 0 as λ rises from the lowest to the highest likelihood. Writing $\mathcal{L}(X)$ for the inverse of this function, the evidence becomes

$$\mathcal{Z} = \int_0^1 \mathcal{L}(X)\,\mathrm{d}X. \tag{2.5}$$

The algorithm (Algorithm 1) evaluates this integral with a set of K live points drawn from the prior. At each iteration i, the live point with the lowest likelihood, $\mathcal{L}_i$, is recorded as a dead point and replaced by a new point drawn from the prior subject to $\mathcal{L} > \mathcal{L}_i$. The live points therefore always sample the prior inside a shrinking likelihood contour. At each step the enclosed volume shrinks by a factor t_i distributed as the largest of K uniform random numbers, with $\langle \ln t_i \rangle = -1/K$, so after i iterations $X_i \approx e^{-i/K}$. The evidence is then the sum

$$\mathcal{Z} \approx \sum_i \mathcal{L}_i\, w_i, \qquad w_i = X_{i-1} - X_i, \tag{2.6}$$

and each dead point becomes a posterior sample with weight $p_i = \mathcal{L}_i w_i / \mathcal{Z}$. The run stops once the live points can add only a small fraction of the evidence, and the remaining live points are then added with the final volume divided equally among them. The statistical uncertainty of $\ln \mathcal{Z}$ is approximately $\sqrt{H/K}$, where

$$H = \int P(\boldsymbol{\theta} \mid D)\, \ln \frac{P(\boldsymbol{\theta} \mid D)}{\pi(\boldsymbol{\theta})}\,\mathrm{d}\boldsymbol{\theta} \tag{2.7}$$

is the information gained from prior to posterior; the number of iterations needed grows as KH.

The difficult step is drawing a new point from the prior inside the current contour. Drawing from the whole prior and rejecting points below the threshold succeeds with probability X_i,

Algorithm 1 Nested sampling

Draw K live points from the prior; set $X_0 = 1$, $\mathcal{Z} = 0$ and $i = 0$
while the live points can still add more than a small fraction of $\mathcal{Z}$ **do**
  $i \leftarrow i + 1$
  Find the live point with the lowest likelihood, $\mathcal{L}_i$
  Set $X_i = e^{-i/K}$ and $w_i = X_{i-1} - X_i$, and add $\mathcal{L}_i w_i$ to $\mathcal{Z}$
  Move that point to the dead points with weight $\mathcal{L}_i w_i$
  Replace it with a new point drawn from the prior subject to $\mathcal{L} > \mathcal{L}_i$
end while
Add the remaining live points to $\mathcal{Z}$ and to the dead points, each with $w = X_i/K$
Divide all weights by $\mathcal{Z}$ to obtain the posterior samples

which falls exponentially with i. Practical samplers therefore restrict the draws to a region that encloses the live points, for example a union of ellipsoids fitted to them, and reject only the draws inside that region that fall below the threshold (Feroz et al., 2019).

2.2.3 Importance nested sampling with Nautilus

Classic nested sampling keeps only the dead points; the many likelihood evaluations rejected while searching for each new live point are discarded. Importance nested sampling (Feroz et al., 2019) instead keeps every point at which the likelihood was evaluated and weights it by the ratio of the posterior to the density from which it was drawn. We use `Nautilus` (Lange, 2023), version 1.0.6, which builds on this idea and replaces the ellipsoidal sampling regions with bounds learned by neural networks.

Nautilus constructs a sequence of nested bounds in the unit cube $[0, 1]^d$ on which the prior is defined (Section 2.2.4). The first bound is the whole cube. Each later bound is built from all points evaluated so far. Its likelihood threshold is the n_{live}-th highest likelihood among them; the points above the threshold are enclosed by one or more ellipsoids; and within each ellipsoid an ensemble of neural networks is trained to predict the rank of a point's likelihood. A point belongs to the bound if it lies inside an ellipsoid and the networks predict a likelihood above the threshold, which lets the bound follow curved or elongated contours more closely than an ellipsoid alone. A new bound is kept only if its volume is smaller than that of the previous one. The region inside one bound but outside all later bounds is a shell, and new points are drawn uniformly within a shell.

Table 2.3: Nautilus settings and run statistics.

	Hubble	TMA
Parameters d	5	10
Live points n_{live}	2000	2000
Neural networks per bound	4	4
Exploration stops at f_{live}	0.01	0.01
Target N_{eff}	10^4	10^4
Bounds built	17	38
Ellipsoids in the final bound	1	1
Likelihood evaluations	44 000	93 200
Final N_{eff}	14 250	20 373
N_{eff} per likelihood evaluation	0.32	0.22
$\ln \mathcal{Z}$	+62.3	+475.5

Sampling proceeds in two phases. In the exploration phase, Nautilus draws points in the innermost shell and adds a new bound after every n_{live} new points above the current threshold (or every $10\,n_{\mathrm{live}}$ likelihood calls), until the n_{live} highest-likelihood points hold less than a fraction f_{live} of the estimated evidence. In the sampling phase, it adds batches of points to the shells that limit the precision of the weighted posterior, until the effective sample size

$$N_{\mathrm{eff}} = \frac{\left(\sum_k w_k\right)^2}{\sum_k w_k^2} \tag{2.8}$$

reaches a target. A point $\boldsymbol{\theta}_k$ in shell j receives the weight $w_k \propto \mathcal{L}(\boldsymbol{\theta}_k)\, V_j / n_j$, where V_j is the estimated volume of the shell and n_j is the number of points in it. The evidence is $\mathcal{Z} = \sum_j V_j\, \bar{\mathcal{L}}_j$, the sum over shells of volume times mean likelihood. Because every evaluated point is used, N_{eff} is a substantial fraction of the number of likelihood calls.

Both runs used the default settings of Nautilus 1.0.6 (Table 2.3) without a fixed random seed, so repeated runs give statistically equivalent but not identical results. The points drawn during the exploration phase are kept, which is also the default; the Nautilus documentation notes that discarding them is required for a fully unbiased posterior and evidence, at the cost of fewer samples.

2.2.4 Prior and likelihood

Nautilus samples the unit cube, and a prior transform maps $\mathbf{u} \in [0,1]^d$ to the physical parameters. Both studies use independent uniform priors, $\theta_k = a_k(2u_k - 1)$, a box of half-width a_k centered on the nominal position; the half-widths are given in Sections 3.1 and 4.1. The function passed to the sampler returns $\ln \mathcal{L}(\boldsymbol{\theta})$ only, since the prior density is uniform inside the box and zero outside. The Hubble transform also rounds each parameter to two decimal places, quantizing translations to $0.01\,\mu$m and tilts to $0.01°$. Within the $\pm 0.167°$ prior, each tilt can therefore take only 35 values, which produces the striped tilt panels of Figure 3.3. The TMA transform does not round.

The observations enter through the basis coefficients. At each field $f \in \mathcal{F}$, the coefficients $\mathbf{c}_f(\boldsymbol{\theta})$ predicted by the forward model are compared with the observed coefficients $\mathbf{c}_{f,\mathrm{obs}}$ through a multivariate Gaussian, and the log-likelihood sums the fields, treating them as independent. For the TMA, $\mathcal{F} = \{(0, 0.5965), (1,0), (0,1), (1,1), (0,-1), (-1,0), (1,-1), (-1,1), (-1,-1), (0.5, 0.5)\}$, where the near-axis field $(0, 0.5965)$ is used in place of $(0,0)$; the Hubble field set is given in Section 3.1. The log-likelihood is

$$\begin{aligned} \ln \mathcal{L}(\boldsymbol{\theta}) = \sum_{f \in \mathcal{F}} \Big\{ & -\frac{1}{2} \left[\mathbf{c}_f(\boldsymbol{\theta}) - \mathbf{c}_{f,\mathrm{obs}}\right]^\top \boldsymbol{\Sigma}_f^{-1} \left[\mathbf{c}_f(\boldsymbol{\theta}) - \mathbf{c}_{f,\mathrm{obs}}\right] \\ & - \frac{N_z}{2} \ln(2\pi) - \frac{1}{2} \ln \det \boldsymbol{\Sigma}_f \Big\} \end{aligned} \tag{2.9}$$

where N_z is the number of retained modes (15 for Hubble, 11 for the TMA) and $\boldsymbol{\Sigma}_f$ is the covariance at field f. Each $\boldsymbol{\Sigma}_f$ is estimated once, before sampling, from 10^4 noise-free forward-model evaluations at misalignments drawn from zero-mean Gaussians, whose widths are given in Sections 3.1 and 4.1. It therefore describes how much the coefficients vary across plausible misalignments, not the measurement noise, which is far smaller (Chapter 5). Because $\boldsymbol{\Sigma}_f$ does not depend on $\boldsymbol{\theta}$, the last two terms of equation (2.9) are constant. They do not affect the posterior, but they set the value of $\ln \mathcal{Z}$, and they are included in the values of Table 2.3.

2.2.5 Using the posterior

Nautilus returns the evaluated points together with their weights. The corner plots (Figures 3.3 and 4.3) show the one- and two-dimensional marginals of these weighted samples, and their panel titles give the weighted median and the 16th–84th percentile range of each parameter. Nautilus stores the points shell by shell, from the unit cube inward, so the final 100 samples are the last points stored in the innermost, highest-likelihood shell. Within one shell every point has the same volume factor, so the weights of these samples are proportional to their likelihoods. They provide the starting point and the bounds for differential evolution (Section 2.3).

2.3 Differential evolution

2.3.1 Why a second stage

The nested-sampling posterior is broad. Its width is set by the calibration covariances $\mathbf{\Sigma}_f$, which describe the spread of the coefficients across the prior rather than the measurement noise (Chapter 5), and some parameters remain nearly unconstrained along degenerate directions. To obtain a precise alignment estimate, we follow nested sampling with a direct minimization of the coefficient residuals,

$$S(\boldsymbol{\theta}) = \sum_{f \in \mathcal{F}} \sum_{k=1}^{N_z} \left[c_{f,k}(\boldsymbol{\theta}) - c_{f,k,\mathrm{obs}} \right]^2 , \tag{2.10}$$

the unweighted sum of squared differences between the forward-model and observed coefficients over all fields. Without noise, S vanishes at the true alignment; with noise, its minimum is offset from the truth by an amount set by the noise (Section 5.3). S does not use $\mathbf{\Sigma}_f$, so the estimate does not depend on the calibration. The cost surface inherits the structure of the likelihood: it can have several local minima and long, flat valleys along degenerate combinations of mirror motions, where a gradient-based optimizer started from a single point can stall. We therefore use differential evolution (DE), a population-based global optimizer introduced by Storn and Price (1997), as implemented in `scipy.optimize.differential_evolution` (Virtanen et al., 2020). DE needs only evalua-

tions of S, not its gradient, which suits a ray-trace forward model whose derivatives are not available analytically.

2.3.2 The algorithm

DE evolves a population of P candidate vectors inside box bounds (Algorithm 2). SciPy sets P to a multiple, `popsize`, of the number of parameters d; with the default `popsize=15`, the population has 75 members for Hubble and 150 for the TMA. Internally, each member is stored in coordinates scaled to $[0, 1]$ across the bounds, so mutation steps are proportional to the bound widths. The population is initialized by Latin hypercube sampling of the bounds, which spreads the members evenly along every parameter, and the starting point $\mathbf{x}_0$ from nested sampling replaces the first member.

Each generation then visits every member $\boldsymbol{\theta}_i$ in turn:

1. **Mutation.** A mutant vector is formed from the current best member and the difference of two other members chosen at random,

$$\mathbf{v}_i = \boldsymbol{\theta}_{\text{best}} + F\,(\boldsymbol{\theta}_{r_1} - \boldsymbol{\theta}_{r_2}). \tag{2.11}$$

   This is the `best1` strategy. The difference vector adapts the step size to the spread of the population: steps are large while the members are scattered and shrink as they converge. The mutation factor F is redrawn uniformly from $[0.5, 1)$ at every generation, a variant known as dithering that can speed up convergence.

2. **Crossover.** A trial vector $\mathbf{u}_i$ takes each parameter from $\mathbf{v}_i$ with probability $C_r = 0.7$ and otherwise keeps it from $\boldsymbol{\theta}_i$; one randomly chosen parameter always comes from $\mathbf{v}_i$. This is binomial crossover, the `bin` in `best1bin`. Parameters that fall outside the bounds are replaced by random values inside them.

3. **Selection.** $S(\mathbf{u}_i)$ is evaluated, and $\mathbf{u}_i$ replaces $\boldsymbol{\theta}_i$ if $S(\mathbf{u}_i) < S(\boldsymbol{\theta}_i)$. If it is also better than the current best member, it becomes the new best.

Because a member is only ever replaced by a better one, the best cost never increases from one generation to the next. The population contracts toward low values of S, while the

random differences between members keep exploring the box. With immediate updating (the Hubble run), a new best member is used at once by the rest of the generation. With deferred updating (the TMA run), the population is updated once per generation, which allows all trial vectors of a generation to be evaluated in parallel.

Algorithm 2 Differential evolution with the best1bin strategy

Initialize P members in the bounds by Latin hypercube sampling; replace the first with $\mathbf{x}_0$
Evaluate S for every member and find the best, $\boldsymbol{\theta}_{\text{best}}$
for each generation, up to the maximum **do**
    Draw F uniformly from $[0.5, 1)$
    for each member $\boldsymbol{\theta}_i$ **do**
        Pick two other members $r_1 \neq r_2$ at random; set $\mathbf{v}_i = \boldsymbol{\theta}_{\text{best}} + F(\boldsymbol{\theta}_{r_1} - \boldsymbol{\theta}_{r_2})$
        Build $\mathbf{u}_i$ from $\mathbf{v}_i$ and $\boldsymbol{\theta}_i$ by binomial crossover with probability C_r
        Replace parameters of $\mathbf{u}_i$ outside the bounds with random values inside them
        if $S(\mathbf{u}_i) < S(\boldsymbol{\theta}_i)$ **then**
            Replace $\boldsymbol{\theta}_i$ by $\mathbf{u}_i$, and $\boldsymbol{\theta}_{\text{best}}$ too if $\mathbf{u}_i$ improves on it
        end if
    end for
    Stop if the spread of the population costs is below the tolerance
end for
Polish $\boldsymbol{\theta}_{\text{best}}$ with L-BFGS-B, and keep the result only if it lowers S

SciPy stops when the standard deviation of the population costs falls below $\text{atol} + \text{tol}\,|\bar{S}|$, where $\bar{S}$ is their mean and $\text{atol} = 0$ by default, or after a maximum number of generations. We used $\text{tol} = 10^{-12}$ for Hubble and 10^{-10} for the TMA. Because the noise sets a floor on S that every member approaches only slowly, the first condition is not reached in practice, and both runs stopped at their limit of 2000 generations. A run of G generations evaluates S $(G + 1)P$ times, plus the evaluations used for polishing: about 1.5×10^5 for Hubble and 3.0×10^5 for the TMA, roughly three times the number of likelihood evaluations used by nested sampling (Table 2.3). Each evaluation of S, like each likelihood evaluation, requires one PSF per field. A callback records the best member and its cost after every generation; these logs give the convergence histories in Figures 3.4 and 4.4. Table 2.4 lists the settings of both runs.

Table 2.4: Differential-evolution settings.

	Hubble	TMA
Parameters d	5	10
Population P (`popsize` $\times\, d$)	75	150
Strategy	`best1bin`	`best1bin`
Mutation factor F	uniform in $[0.5, 1)$	uniform in $[0.5, 1)$
Crossover probability C_r	0.7	0.7
Initialization	Latin hypercube, with $\mathbf{x}_0$	Latin hypercube, with $\mathbf{x}_0$
Updating	immediate	deferred (parallel)
Tolerance `tol`	10^{-12}	10^{-10}
Maximum generations	2000	2000
Evaluations of S before polishing	150 075	300 150
Polishing	L-BFGS-B	L-BFGS-B

2.3.3 Initialization from nested sampling

DE starts from the nested-sampling posterior rather than from the prior. The starting point $\mathbf{x}_0$ is the weighted median of the final 100 posterior samples, which lie in the innermost, highest-likelihood shell (Section 2.2), and each parameter is bounded by the 2.1% and 97.9% weighted quantiles of the same samples. For a Gaussian, these quantiles lie about two standard deviations from the mean, so the box keeps the high-likelihood region while discarding most of the prior. For example, the Hubble despace is searched over 36–43 μm instead of $\pm 200\,\mu$m, and the TMA secondary tilts over 0.007–0.039° and 0.001–0.021° instead of $\pm 0.4°$. Parameters that the posterior does not constrain keep wide bounds, such as the Hubble decenters *sdx* and *sdy* (about 330 and 380 μm wide). DE cannot leave the box, so the box must contain the true alignment; in both runs it did. This starting point is distinct from the full-posterior marginal medians shown in the corner plots (Figures 3.3 and 4.3).

2.3.4 Polishing

After the last generation, SciPy refines the best member with L-BFGS-B, a bounded quasi-Newton method, through `scipy.optimize.minimize`, with gradients estimated by finite differences. The polished point is accepted only if it lowers S, the optimizer reports success and the point lies within the bounds. In both runs, polishing left the solution unchanged to at least nine significant figures. This is weak evidence that differential evolution had converged.

SciPy estimates the gradients for polishing by forward differences with an absolute step of 10^{-8} in each parameter, but numerical noise in the forward model changes S by about 10^{-10} even under parameter changes of 10^{-12}, so near the minimum these estimates are dominated by the noise. In a separate test on the Hubble benchmark, L-BFGS-B with these gradients stopped at least 17 noise standard deviations from the solution from each of eight starting points, whereas central differences with a larger step reached it. The evidence that both runs reached the minimum is instead that their objectives ended at the noise floor predicted in Section 5.3.

2.4 Software and reproducibility

All results were computed with Python 3.11, Optiland 0.6.0, Nautilus 1.0.6, SciPy 1.16.2 and NumPy 2.1.3. The version of Optiland matters: version 0.6.3 refuses the paraxial calculation of the exit pupil, which places the reference sphere for the wavefront, for decentered or tilted mirrors, so it cannot evaluate the misaligned telescopes studied here without modification. Nautilus, differential evolution and the noise added to the observations of the two benchmarks used no fixed random seed, so those observations cannot be regenerated exactly; the conclusions of Chapter 5 depend on the noise level, not on the particular realization. The runs with known moves of the secondary (Section 5.5) used a fixed seed for the noise. The notebooks used for the two benchmarks evaluated the constant term of equation (2.9) with 5 and 10 modes instead of $N_z = 15$ and 11, so they report $\ln \mathcal{Z} = +108.3$ for Hubble and $+484.7$ for the TMA; the values in Table 2.3 are corrected. Because the term is constant, the posteriors and all other results are unaffected. The code and notebooks are available at https://github.com/saikrishanthpm/observatory_psf_alignment.

Chapter 3

Hubble design specifics and results

3.1 Model and setup

As an example of a 2-mirror system, we model a Ritchey–Chrétien-style Hubble Space Telescope in Optiland (Figure 3.2; EPD = 2400 mm, $\lambda = 0.55\,\mu$m) with misalignments applied only to the secondary mirror. The five degrees of freedom are: decenter *sdx*, *sdy* and despace *sdz* (microns), and tilts *stx*, *sty* (degree). Normalized field coordinates are relative to a maximum field angle of 0.038°.

PSFs are computed with `MMDFTPSF` using the default `num_rays=128`, which samples the pupil with 64×64 rays and produces a 256×256 image (Table 2.2); the central 64×64 pixels are retained and projected onto 15 circular-Bessel basis functions (Figure 3.1). The synthetic observations are computed from PSFs with additive Gaussian noise ($\sigma = 0.01$ per pixel, in PSF intensity units, or 0.01% of the ideal peak), while forward-model PSFs in the likelihood are noise free. Synthetic truth coefficients are stored at five normalized fields: $(0, 0)$, $(1, 0)$, $(0, 1)$, $(1, 1)$, and $(-1, -1)$. For each field, a 15×15 covariance matrix is estimated from 10^4 noise-free Monte Carlo draws of $\boldsymbol{\theta}$ under zero-mean Gaussian perturbations ($\sigma = 100\,\mu$m on translations, $\sigma = 0.167$ degrees on tilts), with each draw rounded to two decimal places.

Uniform priors on the unit cube map to $\pm 200\,\mu$m on (sdx, sdy, sdz) and ± 0.167 degrees on (stx, sty); samples from the prior transform are rounded to $0.01\,\mu$m and 0.01 degrees. Nautilus runs with `n_dim=5`. In our completed benchmark, the sampler performed 44 000 likelihood evaluations with $\ln \mathcal{Z} \approx +62.3$ and $N_{\text{eff}} \approx 1.43 \times 10^4$. The weighted median of the last 100 posterior samples provides the starting point for DE.

The DE objective is the unweighted sum of squared coefficient residuals over all five fields, $\sum_{f \in \mathcal{F}} \sum_{k=1}^{15} [c_{f,k}(\boldsymbol{\theta}) - c_{f,k,\text{obs}}]^2$, minimized with `tol=1e-12`; the run reached its limit of 2000

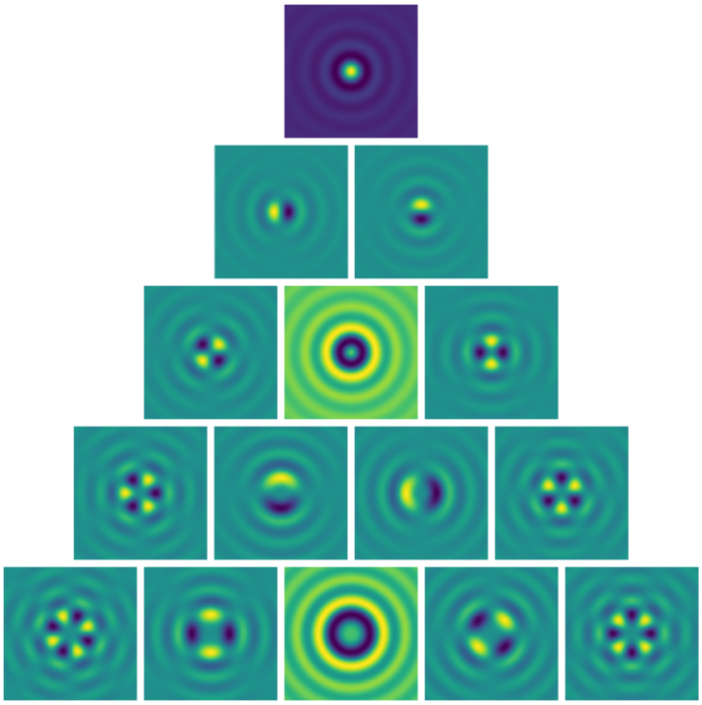


Figure 3.1: The 15 circular-Bessel basis functions used for the Hubble PSFs, $J_{n+1}(2\pi\rho)\cos(m\phi)/\rho$ and $J_{n+1}(2\pi\rho)\sin(m\phi)/\rho$, arranged by radial order $n = 0$–4 from top to bottom. Up to a constant factor, each is the Fourier transform of the Zernike mode in the same position of Figure 4.2 (Noll, 1976). Each panel has its own color scale.

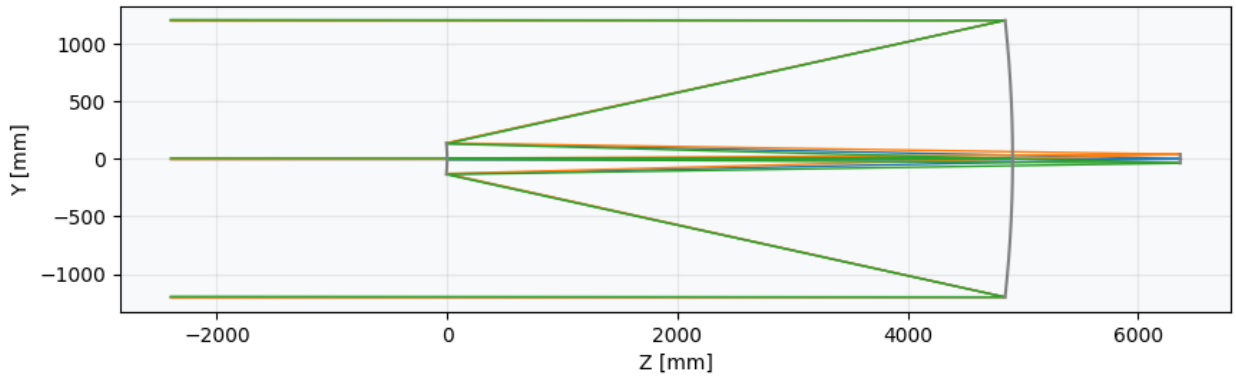


Figure 3.2: Optiland ray trace of the Hubble model. Likelihood evaluation uses circular-Bessel coefficients at five field coordinates (see text).

generations, and L-BFGS-B polishing left the solution unchanged. Nautilus constrains *sdz* and the tilts but leaves *sdx* and *sdy* broad (Section 3.2), so DE must locate the decenters within wide posterior-derived bounds.

3.2 Results

One realization with noisy observations was injected with translations $(sdx, sdy, sdz) = (52.22, -113.86, 38.99)\,\mu$m and tilts $(stx, sty) = (0.13, -0.04)$ degrees. Nautilus ($\ln \mathcal{Z} \approx +62.3$) tightly constrained both tilts and, more loosely, *sdz*, but left *sdx* and *sdy* unconstrained: their marginals in Figure 3.3 are close to flat across the $\pm 200\,\mu$m prior. The weighted median of the final 100 posterior samples, used to start DE, was (51.06, −0.66, 39.60, 0.12, −0.04), and the corresponding 2.1%–97.9% DE bounds span $\approx 327\,\mu$m on *sdx* and $\approx 378\,\mu$m on *sdy*. After 2000 generations, differential evolution reached $\mathbf{c} = (53.77, -110.04, 39.07, 0.1296, -0.0398)$ with objective 6.2×10^{-5}, leaving residuals of 1.55, 3.82 and 0.08 μm on *sdx*, *sdy* and *sdz* and $\lesssim 4 \times 10^{-4}$ degrees on the tilts (Table 3.1). The objective came within 4% of this value by generation 30 and was essentially flat afterwards. The remaining cost is set by the observation noise: the expected sum of squared coefficient noise over the five fields, less the five fitted parameters, is $\approx 5.3 \times 10^{-5}$ (Section 5.3).

With no correction, on-axis OPD RMS is ~ 2.26 waves. Using the DE starting point (the median of the final 100 posterior samples) alone reduces this to ~ 0.014 waves. After DE refinement, on-axis OPD RMS falls to $\sim 1.3 \times 10^{-3}$ waves. Because the objective is limited

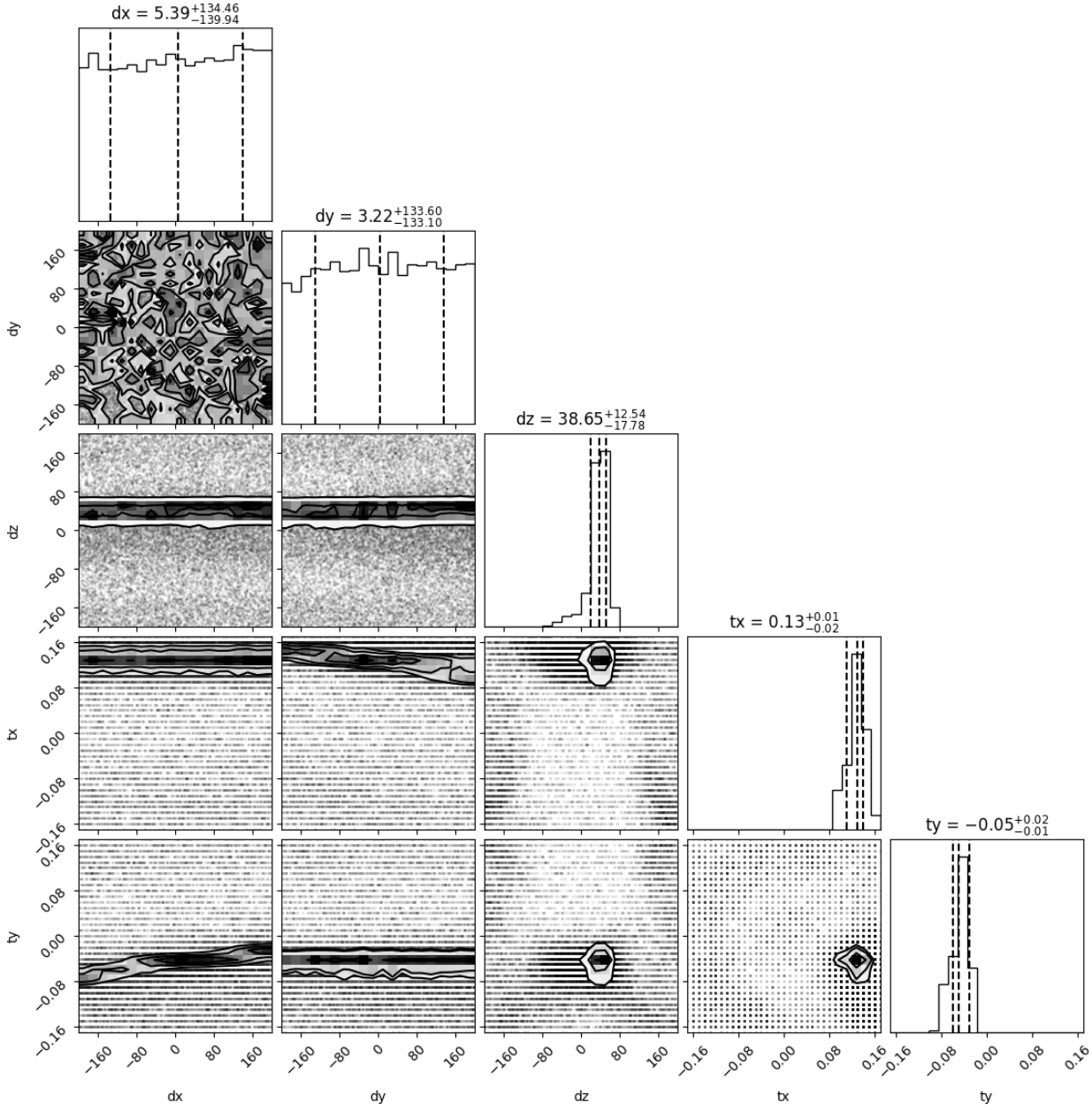


Figure 3.3: Nautilus posterior corner plot for the five secondary-misalignment parameters of the Hubble model (axes labelled dx, dy, dz, tx, ty correspond to sdx, sdy, sdz, stx, sty). Titles give the full-posterior marginal medians.

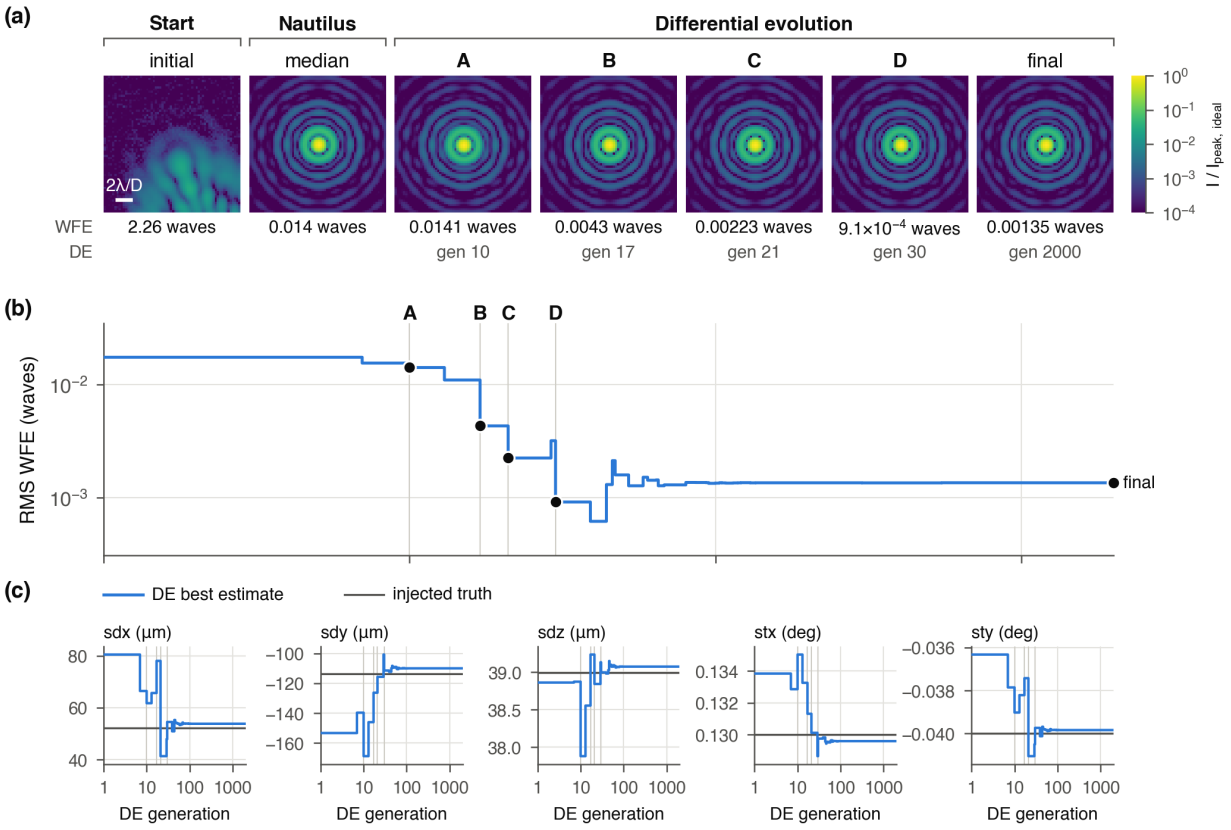


Figure 3.4: Hubble benchmark with noisy PSFs: on-axis PSF and RMS wavefront error (WFE; the on-axis OPD RMS quoted in the text) as the alignment is refined. (a) PSFs for the uncorrected system (the observed PSF, with Gaussian noise of $\sigma = 0.01\%$ of the ideal peak), the Nautilus starting point (median of the final 100 posterior samples), DE solutions at generations 10, 17, 21 and 30 (A–D) and the DE solution after 2000 generations (final). (b) RMS WFE of the DE best estimate versus generation. (c) DE best estimate of each secondary-mirror parameter versus generation, with the injected truth.

Table 3.1: Hubble secondary: injected truth and DE-recovered misalignment (noisy PSFs, $\sigma = 0.01\%$ of the ideal peak; after 2000 DE generations and L-BFGS-B polishing). Tilts are in degrees.

Parameter	True value	Recovered (DE)	Units
sdx	52.22	53.77	μm
sdy	-113.86	-110.04	μm
sdz	38.99	39.07	μm
stx	0.13	0.1296	degree
sty	-0.04	-0.0398	degree

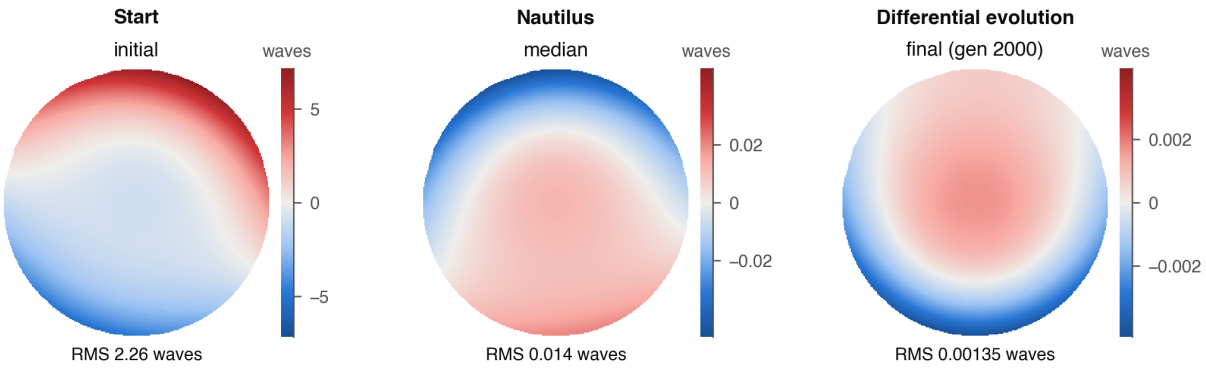


Figure 3.5: Residual on-axis OPD maps of the Hubble model: the difference between the nominal design and the system with residual misalignment $\boldsymbol{\theta}_{\text{true}} - \mathbf{c}$, for the uncorrected system, the Nautilus starting point and the DE solution after 2000 generations. Piston is removed and tilt is retained, as in the quoted RMS WFE. Each panel has its own symmetric color scale in waves, with zero in gray. By variance, the final residual is 57% defocus and 39% tilt.

by noise, the on-axis OPD RMS does not decrease monotonically: it reaches 6×10^{-4} waves at generation 39 before settling at 1.3×10^{-3} waves. Figure 3.3 shows posterior marginals for all five secondary misalignments, and Figure 3.4 shows the on-axis PSF and RMS wavefront error at each stage of the refinement. The residual OPD maps (Figure 3.5) show what remains: the final residual is mostly defocus (57% of the variance) and tilt (39%), the low-order aberrations left by the small despace error and by the decenter–tilt trade-off described in Section 5.3.

Chapter 4

TMA design specifics and results

4.1 Model and setup

We apply the same inference pipeline to a compact three-mirror anastigmat (TMA) based on the optical design described in Kim et al. (2023), implemented in Optiland as a four-mirror prescription (Figure 4.1) with entrance-pupil diameter EPD = 6471.26 mm and monochromatic wavelength $\lambda = 0.65\,\mu$m. Unlike the Hubble case, misalignments are applied to both the primary and secondary mirrors: ten rigid-body degrees of freedom parameterize decenter (x, y), despace (z), and tilt (x, y) on each mirror. Translations are specified in microns and tilts are in degrees. Normalized field coordinates are relative to a maximum field angle of 0.285°.

For each trial alignment vector $\boldsymbol{\theta}$, we build the TMA model and propagate the field with `MMDFTPSF` using 512 rays, and retain a central 256×256 pixel intensity patch. The patch is projected onto the first 11 Zernike modes (Figure 4.2), yielding an 11-element coefficient vector $\mathbf{c}_f(\boldsymbol{\theta})$ at each field point f. The synthetic observations $\mathbf{c}_{f,\mathrm{obs}}$ are computed from PSFs with additive Gaussian noise ($\sigma = 0.01$ per pixel, in PSF intensity units), while forward-model PSFs in the likelihood are noise free. Nested sampling and differential evolution used separate noise realizations at the same σ, which has no practical effect on the nested-sampling stage (Section 5.2). Each covariance $\boldsymbol{\Sigma}_f$ is estimated from 10^4 noise-free Monte Carlo draws with $\sigma = 250\,\mu$m on (pdx, pdy), $50\,\mu$m on pdz, $800\,\mu$m on (sdx, sdy), $100\,\mu$m on sdz, 0.003 degrees on (ptx, pty), and 0.13 degrees on (stx, sty), with each draw rounded to two decimal places. The rounding sets about 90% of the primary-tilt draws to exactly zero, but their standard deviation, 0.0031 degrees, stays close to the intended value.

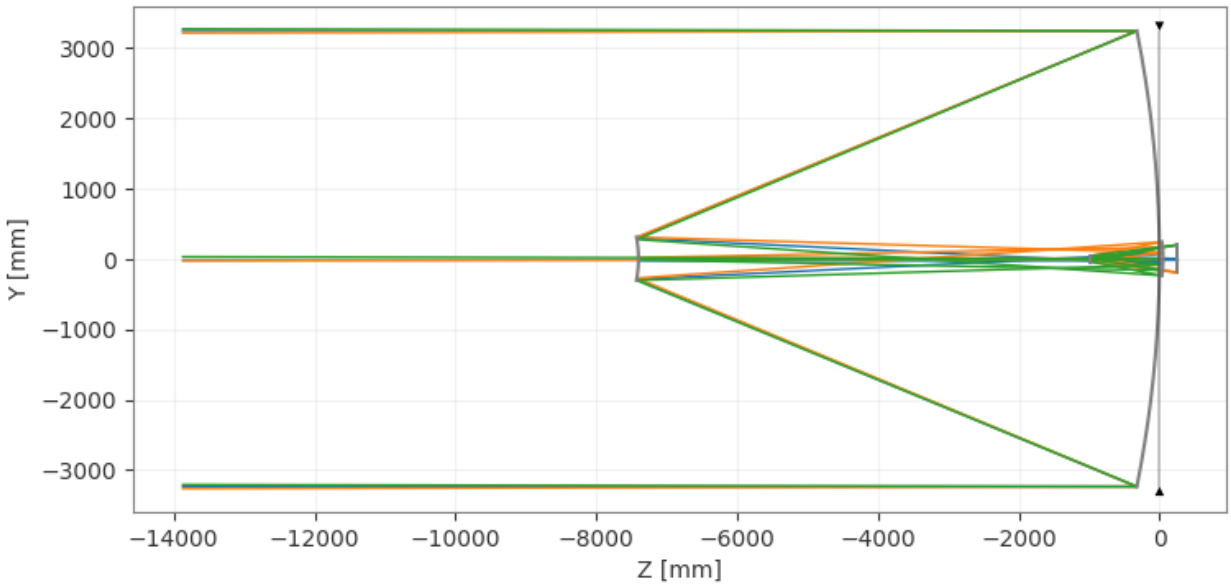


Figure 4.1: Raytrace of the TMA model in Optiland. Multi-field PSF features used in inference are evaluated at ten field coordinates across the focal plane (see text).

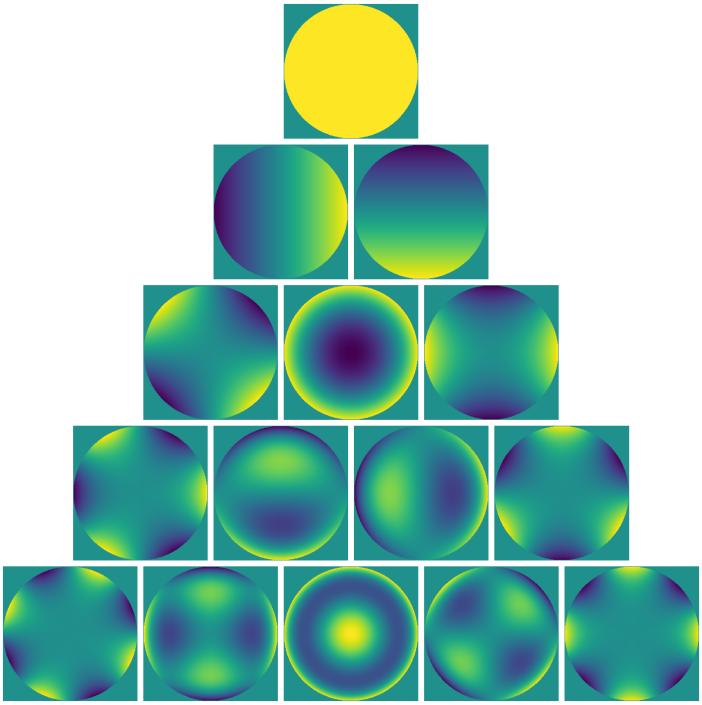


Figure 4.2: The first 15 Zernike modes in Noll's ordering (Noll, 1976), arranged as in Figure 3.1 so that each mode sits where its Fourier transform does. The modes are defined on the disk inscribed in the 256×256 image patch and are zero outside it. The TMA PSFs are projected onto the first 11: rows $n = 0$–3 and primary spherical aberration at the center of the last row. Each panel has its own symmetric color scale, so zero has the same color in every panel.

The prior transform maps uniform unit-cube samples to independent uniform boxes: primary decenters (pdx, pdy) and despace $pdz \in [-800, 800]\,\mu$m, secondary decenters (sdx, sdy) and despace $sdz \in [-2500, 2500]\,\mu$m, primary tilts $\in [-0.01, 0.01]$ degrees, and secondary tilts $\in [-0.4, 0.4]$ degrees. The weighted median $\mathbf{x}_0$ and the 2.1%–97.9% weighted quantiles of the final 100 posterior samples initialize and bound differential evolution. The DE objective is the unweighted sum of squared coefficient residuals over all ten fields, $\sum_{f\in\mathcal{F}}\sum_{k=1}^{11}[c_{f,k}(\boldsymbol{\theta}) - c_{f,k,\mathrm{obs}}]^2$, minimized with `tol=1e-10`; the run reached the limit of 2000 generations (final objective 2.2×10^{-7}) before L-BFGS-B polishing, which left the solution unchanged.

Recovered alignments $\mathbf{c}$ are validated by constructing a residual telescope with misalignments $(\boldsymbol{\theta}_{\mathrm{true}} - \mathbf{c})$ and comparing on-axis PSF images and OPD maps against the nominal system via `OPD` at field $(0, 0)$. We report both per-component residuals $\boldsymbol{\theta}_{\mathrm{true}} - \mathbf{c}$ and the RMS of the OPD difference in waves.

Both studies use Optiland for ray tracing, Nautilus for nested sampling, and SciPy (Virtanen et al., 2020) differential evolution for final refinement; the TMA study uses POPPY (Perrin et al., 2012) for the Zernike basis.

4.2 Results

The TMA study uses the noisy observations described in Sec. 4.1. Ten-dimensional nested sampling is computationally expensive; posterior medians need not coincide with injected truth when the likelihood contains flat directions. The posterior marginals in Figure 4.3 suggest partial degeneracies in the ten-dimensional problem. The primary decenter *pdx* and both primary tilts are close to flat across their prior ranges, the secondary decenters remain broad, and several pairs are strongly correlated, notably *pdz*–*sdz*, *ptx*–*sdy* and *pty*–*sdx*. The tilt–decenter correlations have a familiar optical origin: a tilt of one mirror about the x (y) axis and a decenter of the other along y (x) both introduce coma that is nearly constant across the field, so one can largely cancel the coma introduced by the other. The same mechanism made secondary-mirror misalignment on JWST detectable only through field-dependent aberrations (Chapter 1): a compensating motion of another mirror cancels the field-constant error and leaves a field-dependent residual, which is why our likelihood

Table 4.1: TMA: injected truth and DE-recovered alignment (after 2000 generations and L-BFGS-B polishing).

Parameter	True value	Recovered	Units
pdx	-152.75	-155.30	μm
pdy	316.91	327.15	μm
pdz	26.63	30.26	μm
sdx	-42.31	3.81	μm
sdy	923.33	932.86	μm
sdz	-32.75	-29.22	μm
ptx	2.98×10^{-5}	6.71×10^{-6}	degree
pty	1.61×10^{-5}	-3.50×10^{-4}	degree
stx	0.0269	0.0268	degree
sty	0.0098	0.0095	degree

combines several field points. The *pdz*–*sdz* correlation similarly reflects the trade-off between the two despaces in setting focus. Only the secondary tilts are tightly constrained. Distinct alignment states may therefore produce similar Zernike vectors, and the likelihood alone may not determine every parameter uniquely.

Nautilus performed 93 200 likelihood evaluations with $\ln \mathcal{Z} \approx +475.5$ and $N_{\text{eff}} \approx 2.04 \times 10^4$. With no correction, on-axis OPD RMS is ~ 8.24 waves, and the DE starting point $\mathbf{x}_0$ (the median of the final 100 posterior samples) alone reduces this to ~ 0.99 waves. Table 4.1 reports the truth and DE-refined alignment states: the refinement reduces on-axis OPD RMS to $\sim 2.5 \times 10^{-3}$ waves while leaving residuals of up to $\sim 46\,\mu$m on the translations (largest on secondary decenter *sdx*; the other translations are within $10.3\,\mu$m). Primary tilts are near zero at truth and remain poorly determined relative to translations. Figure 4.4 shows the on-axis PSF and RMS wavefront error at each stage of the refinement, and Figure 4.5 the residual OPD maps. As for Hubble, the final residual is dominated by low-order terms, tilt (66% of the variance) and defocus (26%).

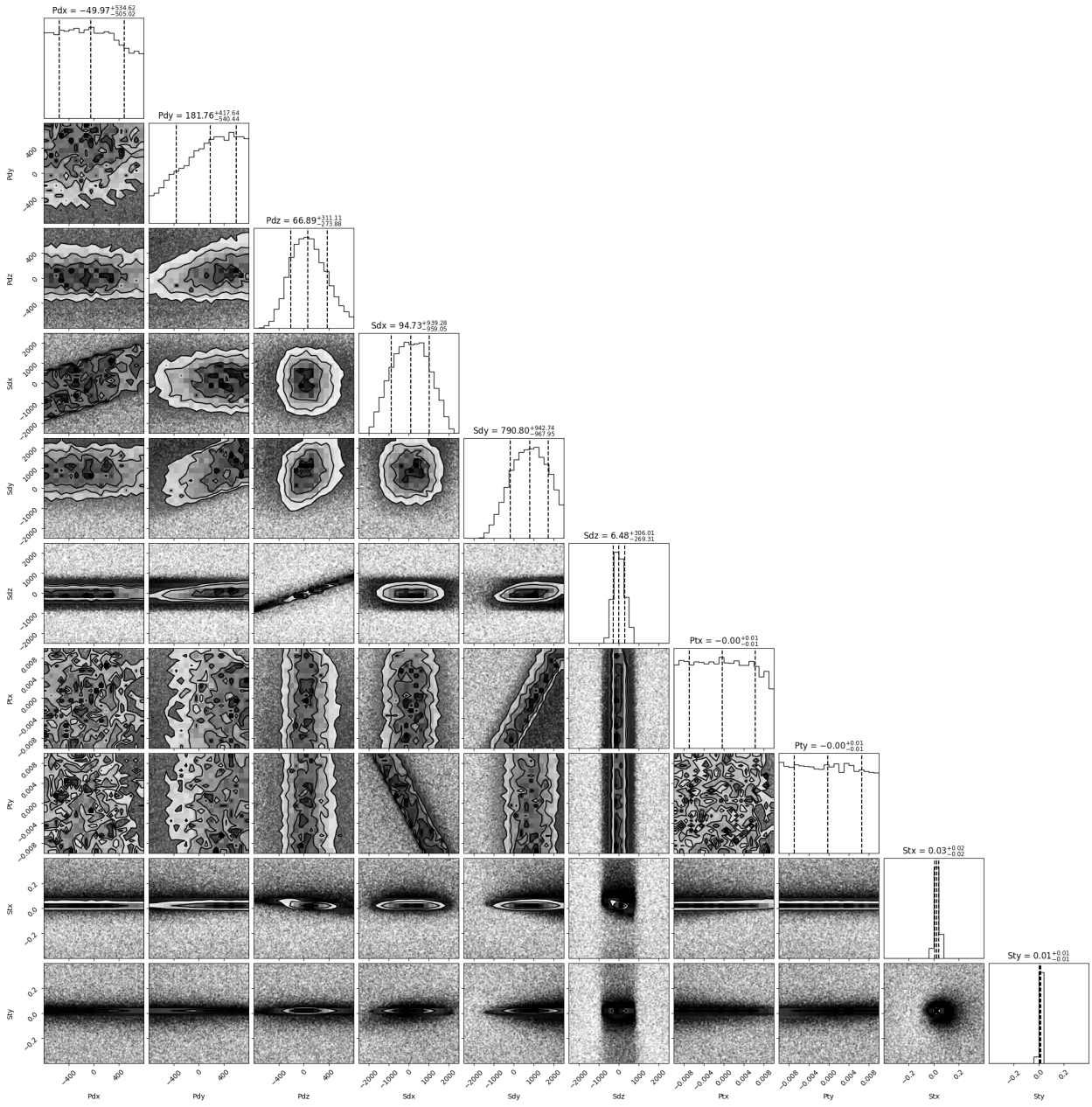


Figure 4.3: Nautilus posterior corner plot for the ten primary/secondary misalignment parameters of the TMA model.

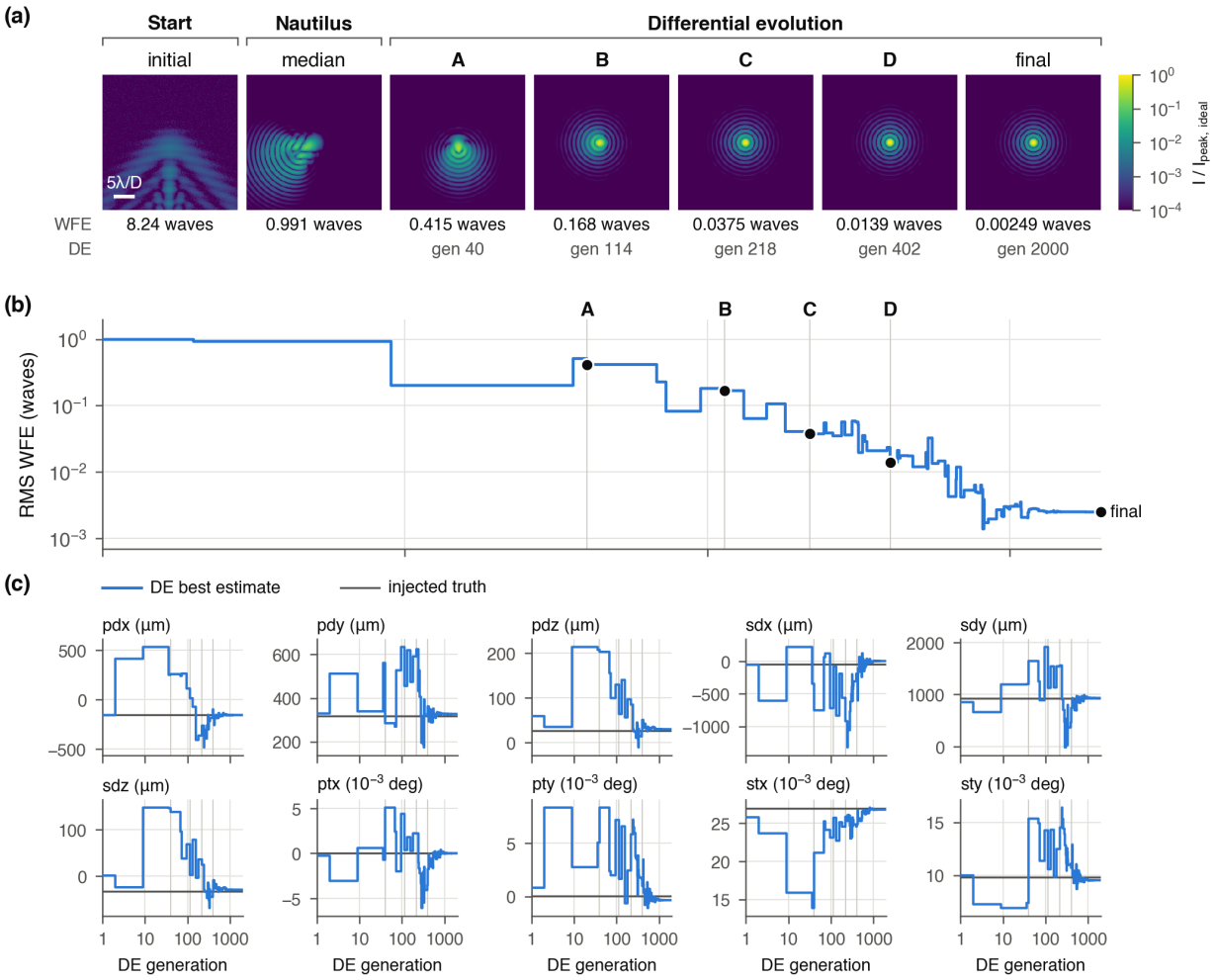


Figure 4.4: TMA benchmark: (a) On-axis PSFs at $\lambda = 0.65\,\mu$m: the observed misaligned PSF (Gaussian noise, $\sigma = 0.01\%$ of the ideal peak), the Nautilus starting point (median of the final 100 posterior samples), during differential evolution (DE) at generations 40, 114, 218 and 402 (A–D), and after 2000 generations (final). Intensity is relative to the aligned system's peak, so each panel's peak equals its Strehl ratio (log stretch, floor 10^{-4}). Values are RMS wavefront error in waves, the RMS of the OPD difference from the nominal design. (b) RMS WFE of the best DE member at each generation; letters mark the PSFs in (a). (c) Best-member value of each alignment parameter at each generation (blue) and the injected truth (gray).

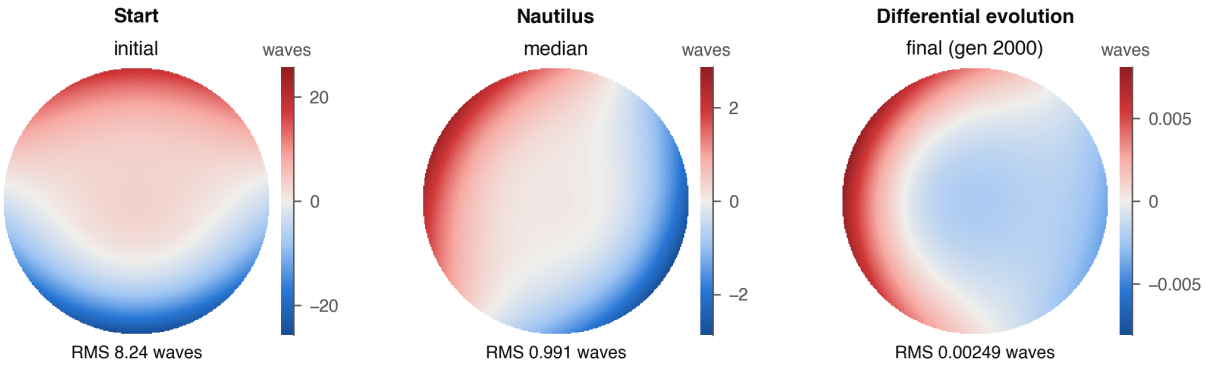


Figure 4.5: Residual on-axis OPD maps of the TMA: the difference between the nominal design and the system with residual misalignment $\boldsymbol{\theta}_{\text{true}} - \mathbf{c}$, for the uncorrected system, the Nautilus starting point and the DE solution after 2000 generations and L-BFGS-B polishing. Piston is removed and tilt is retained, as in the quoted RMS WFE. Each panel has its own symmetric color scale in waves, with zero in gray. By variance, the final residual is 66% tilt and 26% defocus.

Chapter 5

Noise in both models

Both benchmarks infer the alignment from noisy observations: zero-mean Gaussian noise is added to every observed PSF (Chapters 3 and 4), while the forward model and the calibration covariances $\mathbf{\Sigma}_f$ are noise free. This chapter follows that noise through the pipeline: into the basis coefficients (Section 5.1), into the likelihood explored by nested sampling (Section 5.2) and into the differential-evolution estimate (Section 5.3). It then asks how the results change with the noise level (Section 5.4) and how much observations after known moves of the secondary improve the precision (Section 5.5). The calculations linearize the forward model about the injected truth and are checked against the two pipeline runs and against Monte Carlo simulations.

5.1 Noise model and coefficient noise

The observed PSF at each field is $I_{\text{obs}} = I(\boldsymbol{\theta}_{\text{true}}) + n$, where n is independent Gaussian noise with standard deviation $\sigma = 0.01$ in each pixel of the retained image (64×64 pixels for Hubble, 256×256 for the TMA). Optiland normalizes the PSF so that an aligned system peaks at 100, so σ is 10^{-4} of the ideal peak. One noise realization is drawn per field. The noise is additive and independent of the signal, like detector read noise; photon noise, background and other detector effects are not modelled.

Each coefficient is a linear projection of the image onto a basis function B_k,

$$c_k = \frac{\langle B_k, I \rangle}{\langle B_k, B_k \rangle}, \tag{5.1}$$

Table 5.1: Noise in coefficient space, in the likelihood and in the DE objective for $\sigma = 0.01$ (10^{-4} of the ideal peak). Expected values follow from equation (5.2); realized values are from the runs in Chapters 3 and 4.

	Hubble	**TMA**
Retained image (pixels)	64×64	256×256
Basis (modes)	circular-Bessel (15)	Zernike (11)
Fields; coefficients in total	5; 75	10; 110
Coefficient noise s.d. per mode	$(0.3\text{–}1.1) \times 10^{-3}$	4.4×10^{-5}
Largest noise correlation between modes	0.28	< 0.01
Median $N_{kk}/\Sigma_{f,kk}$	1.4×10^{-5}	6.9×10^{-5}
Shift of $-2 \ln \mathcal{L}$ at truth, expected	0.0039	0.022
Shift of $-2 \ln \mathcal{L}$ at truth, realized	0.0039	0.027
DE cost floor, expected	$(5.3 \pm 1.0) \times 10^{-5}$	$(2.0 \pm 0.3) \times 10^{-7}$
DE cost, final	6.2×10^{-5}	2.2×10^{-7}

where $\langle \cdot, \cdot \rangle$ is the sum over pixels. The pixel noise therefore becomes zero-mean Gaussian noise on the coefficients, with covariance

$$N_{jk} = \sigma^2 \, \frac{\langle B_j, B_k \rangle}{\langle B_j, B_j \rangle \, \langle B_k, B_k \rangle}. \tag{5.2}$$

$\mathbf{N}$ is the same at every field and does not depend on $\boldsymbol{\theta}$. For the TMA, the Zernike modes are orthogonal and have equal norm on the unit disk on which they are defined, which covers about 5.1×10^4 of the 256×256 pixels. Every coefficient therefore has the same noise, $\sigma/\sqrt{5.1 \times 10^4} = 4.4 \times 10^{-5}$, and the modes are uncorrelated. The circular-Bessel functions used for Hubble are neither normalized nor orthogonal on the 64×64 grid: the coefficient noise ranges from 3.0×10^{-4} to 1.1×10^{-3} across the 15 modes, with correlations of up to 0.28 between modes. The Hubble coefficients are also noisier because each projection averages over only 4096 pixels. Table 5.1 summarizes these quantities.

5.2 Effect on the likelihood

The calibration covariances $\boldsymbol{\Sigma}_f$ in the likelihood (equation 2.9) come from noise-free Monte Carlo draws of $\boldsymbol{\theta}$, so they describe how much the coefficients vary across the prior, not measurement noise. The noise covariance is tiny by comparison. The diagonal of $\mathbf{N}$ is a

median 1.4×10^{-5} (at most 8.1×10^{-5}) of the diagonal of $\mathbf{\Sigma}_f$ for Hubble, and 6.9×10^{-5} (at most 3.6×10^{-4}) for the TMA.

At the truth, the noise $\mathbf{n}_f$ on the observed coefficients raises $-2 \ln \mathcal{L}$ by $\sum_f \mathbf{n}_f^\top \mathbf{\Sigma}_f^{-1} \mathbf{n}_f$, whose expected value is $\sum_f \mathrm{tr}(\mathbf{\Sigma}_f^{-1} \mathbf{N})$. This is 0.0039 for Hubble and 0.022 for the TMA. Evaluating the likelihood at the truth in the two runs gives 0.0039 and 0.027. Differences in $\ln \mathcal{L}$ that matter to nested sampling are of order unity, so at this noise level the noise has no material effect on the posteriors in Figures 3.3 and 4.3: their widths are set by $\mathbf{\Sigma}_f$. The shift grows as σ^2 and would reach unity at $\sigma \approx 0.16$ for Hubble and $\sigma \approx 0.07$ for the TMA. For the TMA, nested sampling and DE used separate noise realizations at the same σ; by the same argument, this has no practical effect on the nested-sampling stage.

The posterior credible intervals are therefore not noise-limited uncertainties. The Hubble *sdx* and *sdy* marginals are nearly flat across the $\pm 200\,\mu$m prior, while the noise alone limits these parameters to $\approx 4\,\mu$m (Section 5.3). A likelihood built from $\mathbf{N}$ instead of $\mathbf{\Sigma}_f$ would concentrate the posterior around the noise-limited solution, at the cost of a far more sharply peaked likelihood for the sampler to explore. In the present pipeline, nested sampling serves as a global search, and the precision comes from the DE refinement.

5.3 Effect on the differential-evolution estimate

DE minimizes the unweighted sum of squared coefficient residuals over all fields, $S(\boldsymbol{\theta}) = \|\mathbf{c}(\boldsymbol{\theta}) - \mathbf{c}_{\mathrm{obs}}\|^2$, where $\mathbf{c}$ stacks the coefficients of every field (75 for Hubble, 110 for the TMA). Without noise, the minimum is at the truth, where $S = 0$; with noise, it moves. Linearizing the forward model about the truth, $\mathbf{c}(\boldsymbol{\theta}) \approx \mathbf{c}(\boldsymbol{\theta}_{\mathrm{true}}) + \mathbf{J}\,\delta\boldsymbol{\theta}$, the least-squares estimate and its covariance are

$$\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_{\mathrm{true}} \approx \mathbf{A}\,\mathbf{n}, \qquad \mathrm{Cov}(\hat{\boldsymbol{\theta}}) = \mathbf{A}\,\mathbf{N}_{\mathrm{all}}\,\mathbf{A}^\top, \qquad \mathbf{A} = (\mathbf{J}^\top \mathbf{J})^{-1} \mathbf{J}^\top, \tag{5.3}$$

where $\mathbf{n}$ is the stacked coefficient noise and $\mathbf{N}_{\mathrm{all}}$ is block diagonal, with one copy of $\mathbf{N}$ per field. The expected minimum cost is $\mathrm{tr}[(\mathbf{I} - \mathbf{J}\mathbf{A})\,\mathbf{N}_{\mathrm{all}}]$, the noise power left after the fit absorbs the components along the columns of $\mathbf{J}$. We compute $\mathbf{J}$ by central differences at the truth. The steps are $0.2\,\mu$m and 2×10^{-5} degrees for Hubble; for the TMA they are $0.5\,\mu$m,

Table 5.2: Noise-limited precision of the DE estimate, $\sqrt{\operatorname{diag}\operatorname{Cov}(\hat{\boldsymbol{\theta}})}$ from equation (5.3), compared with the residuals $\boldsymbol{\theta}_{\text{true}} - \mathbf{c}$ of the runs (Tables 3.1 and 4.1); z is their ratio. Translations are in μm and tilts in degrees.

Parameter	**Predicted s.d.**	**Residual**	$\boldsymbol{z}$
Hubble			
sdx	3.62	-1.55	-0.43
sdy	3.96	-3.82	-0.97
sdz	0.075	-0.082	-1.09
stx	3.8×10^{-4}	4.1×10^{-4}	1.07
sty	3.6×10^{-4}	-1.5×10^{-4}	-0.42
TMA			
pdx	11.3	2.55	0.23
pdy	8.79	-10.24	-1.16
pdz	2.37	-3.63	-1.53
sdx	50.3	-46.12	-0.92
sdy	48.4	-9.53	-0.20
sdz	2.33	-3.53	-1.52
ptx	3.4×10^{-4}	2.3×10^{-5}	0.07
pty	3.5×10^{-4}	3.7×10^{-4}	1.05
stx	2.7×10^{-4}	1.9×10^{-4}	0.72
sty	2.9×10^{-4}	2.8×10^{-4}	0.95

5×10^{-6} degrees on the primary tilts and 5×10^{-5} degrees on the secondary tilts. Doubling the steps changes $\mathbf{J}$ by less than 1%.

The expected minimum costs are $(5.3 \pm 1.0) \times 10^{-5}$ for Hubble and $(2.0 \pm 0.3) \times 10^{-7}$ for the TMA, where the uncertainty is the scatter between noise realizations. The DE runs ended at 6.2×10^{-5} and 2.2×10^{-7}, within one standard deviation of these values. Both runs therefore reached the noise floor of their objectives. The flat cost after generation ~ 30 in the Hubble run (Section 3.2) and the 2.2×10^{-7} final cost of the TMA run are both set by the noise.

Table 5.2 compares the predicted standard deviations with the residuals of the two runs. Every residual lies within 1.6 standard deviations of zero. Taking the correlations into account, the squared Mahalanobis distance of the residual vector, $\delta\boldsymbol{\theta}^{\top}\operatorname{Cov}(\hat{\boldsymbol{\theta}})^{-1}\delta\boldsymbol{\theta}$, is 3.8 for Hubble and 4.9 for the TMA, typical values for χ^2 distributions with 5 and 10 degrees of freedom. The residuals in both runs are thus what the noise predicts: DE found the least-squares solution, and the remaining error comes from the noise rather than from the

optimizer. The same holds for image quality. Propagating $\mathrm{Cov}(\hat{\boldsymbol{\theta}})$ to the on-axis OPD by Monte Carlo (300 draws) gives a median RMS WFE of 1.2×10^{-3} waves for Hubble (16th–84th percentiles 0.7–1.7 $\times 10^{-3}$) and 4.2×10^{-3} waves for the TMA (2.4–6.9 $\times 10^{-3}$). Both ranges contain the 1.35×10^{-3} and 2.5×10^{-3} waves reached by the runs.

The noise-limited errors are not spread evenly across the parameters: they concentrate along the same near-degeneracies that shape the posteriors (Figure 5.1). For Hubble, the decenter errors are almost perfectly correlated with the secondary tilt about the perpendicular axis ($r = -0.996$ for *sdy*–*stx* and $r = +0.995$ for *sdx*–*sty*). A decenter and a tilt of the secondary both introduce nearly field-constant coma, the mechanism described for the TMA in Section 4.2, so one can largely compensate the other. If all the other parameters were known, the noise would determine *sdx* and *sdy* to 0.36 and 0.32 μm instead of 3.6 and 4.0 μm. For the TMA, the strongest correlations are *pdz*–*sdz* ($r = 1.00$), *pty*–*sty* (0.99), *sdx*–*sty* (-0.98), *sdy*–*ptx* (0.98) and *sdx*–*pty* (-0.98), which include the pairs found in the TMA posterior. If the other nine parameters were known, the noise alone would determine *sdx* to 0.30 μm; the degeneracies amplify this to 50 μm. The large *sdx* residual of the TMA run (46 μm) is therefore noise projected onto a nearly degenerate direction, and the near-unity *pdz*–*sdz* correlation is the focus trade-off between the two despaces.

Because the circular-Bessel coefficient noise is unequal across modes and correlated, weighting the DE objective by $\mathbf{N}^{-1}$ (generalized least squares) would give the minimum-variance estimate. For Hubble, this would reduce the predicted standard deviations by only 5–13%. For the TMA, $\mathbf{N}$ is proportional to the identity, so weighting would change nothing.

The noise-free limit is consistent with this picture. In an earlier noise-free run of the Hubble benchmark, with an on-axis DE objective, DE reached an objective below 10^{-15} and recovered the translations to better than 0.01 μm, with an on-axis OPD RMS of $\sim 3 \times 10^{-8}$ waves.

5.4 Dependence on noise level

In the linear approximation, the parameter errors and the WFE scale in proportion to σ: the values for another noise level follow by multiplying those in Table 5.2 by $\sigma/0.01$. To test where this breaks down, we simulated noisy observations at the truth for $\sigma = 0.01$, 0.03, 0.1

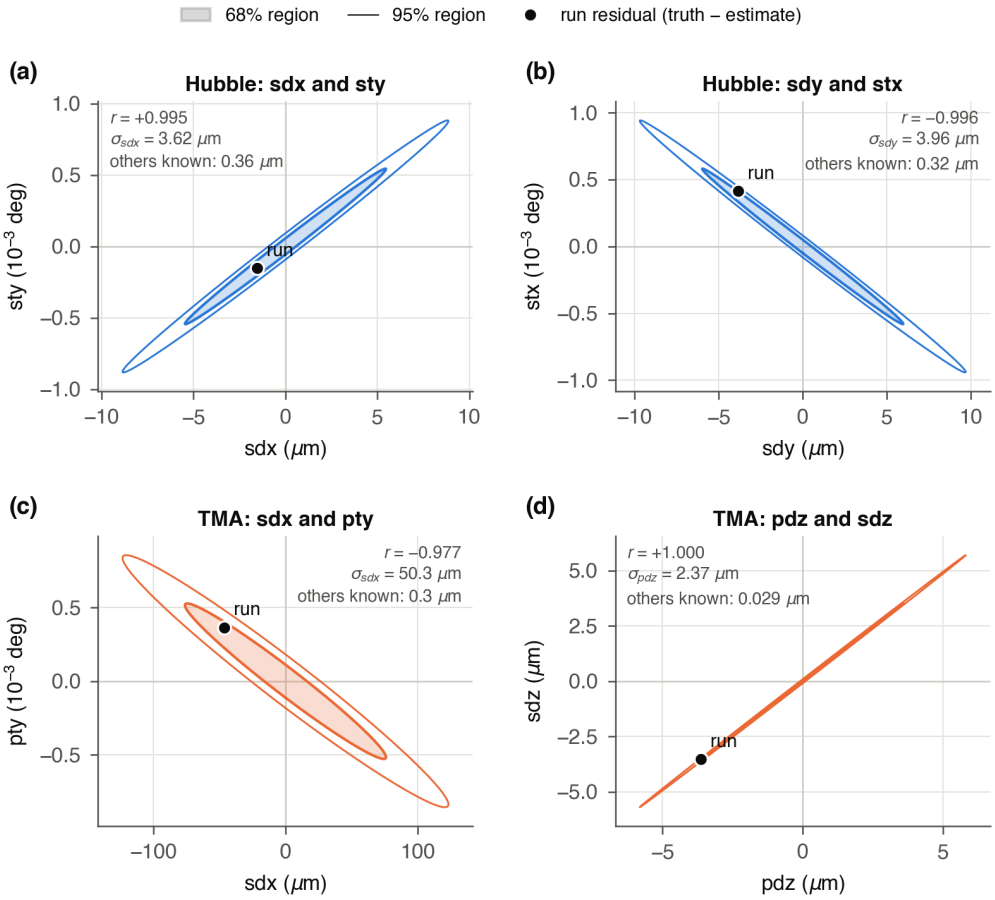


Figure 5.1: Noise-limited scatter of the DE estimate in the planes of the strongest correlations: the 68% and 95% regions of the linearized covariance (equation 5.3) for Hubble (a, b; blue) and the TMA (c, d; orange), with the residual $\boldsymbol{\theta}_{\text{true}} - \mathbf{c}$ of each run (black). Tilts are in 10^{-3} degrees. Each panel gives the correlation coefficient, the standard deviation of the horizontal parameter, and its standard deviation if all other parameters were known.

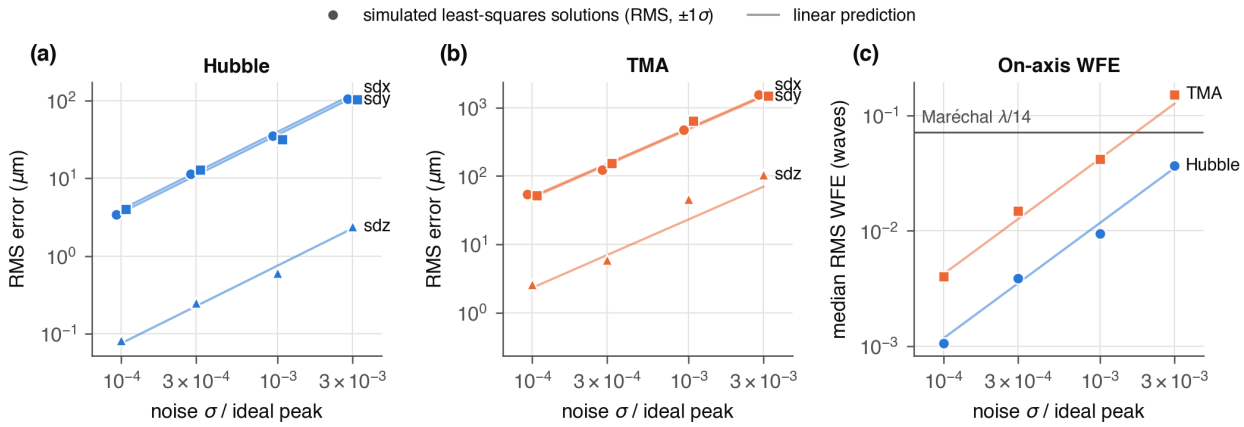


Figure 5.2: Parameter errors and on-axis WFE versus noise level. (a, b) RMS errors of the secondary decenters and despace for Hubble and the TMA. Points are simulated least-squares solutions (50 and 30 noise realizations per level for Hubble and the TMA; error bars show the $\pm 1\sigma$ uncertainty of each RMS value), and lines are the linear prediction, proportional to σ. The *sdx* and *sdy* points are offset slightly along σ so that both are visible. The TMA *pdz* errors (Table 5.3) follow *sdz*. (c) Median on-axis RMS WFE of the same solutions, with the Maréchal criterion $\lambda/14$.

Table 5.3: RMS parameter errors and median on-axis RMS WFE of simulated least-squares solutions at four noise levels, with the linear prediction in parentheses. $\sigma = 0.01$ in Optiland units is 10^{-4} of the ideal peak. With 50 (Hubble) and 30 (TMA) realizations per level, the simulated RMS values are uncertain by about 10% and 13%.

	Hubble			TMA		
σ/I_{peak}	sdx (μm)	sdy (μm)	WFE (10^{-3})	sdx (μm)	pdz (μm)	WFE (10^{-3})
10^{-4}	3.41 (3.62)	4.01 (3.96)	1.06 (1.17)	53.7 (50.3)	2.67 (2.37)	4.02 (4.22)
3×10^{-4}	11.3 (10.9)	12.9 (11.9)	3.89 (3.51)	123 (151)	6.09 (7.12)	14.9 (12.6)
10^{-3}	35.0 (36.2)	31.6 (39.6)	9.49 (11.7)	472 (503)	46.9 (23.7)	41.8 (42.2)
3×10^{-3}	105 (109)	102 (119)	36.7 (35.1)	1550 (1510)	107 (71.2)	152 (126)

and 0.3 (10^{-4} to 3×10^{-3} of the ideal peak), with 50 realizations per level for Hubble and 30 for the TMA. Each least-squares problem was solved from the truth with twelve Gauss–Newton iterations, holding $\mathbf{J}$ at its value at the truth; in spot checks, the final steps were below 10^{-3} of the predicted standard deviation. The simulations start from the truth and ignore the DE bounds, so they describe the least-squares solution nearest the truth rather than every solution DE might return. Table 5.3 and Figure 5.2 compare the RMS errors and the median WFE of these solutions with the linear predictions.

For Hubble, the simulated errors and WFE follow the linear prediction to within about 20% at every level, so the linear scaling holds up to at least $\sigma = 0.3$. The median WFE there, 0.037 waves, is still within the Maréchal criterion of $\lambda/14 \approx 0.071$ waves. For the

TMA, the linear prediction holds for $\sigma \leq 0.03$. At $\sigma = 0.1$, the despace errors are twice the linear prediction (47 μm on *pdz* against 24 μm), as the *pdz*–*sdz* trade-off is no longer linear. At $\sigma = 0.3$, the median WFE of 0.15 waves is 20% above the linear value and exceeds the Maréchal criterion; the linear prediction places the crossing near $\sigma \approx 0.17$, or 0.17% of the ideal peak. At that level, the RMS error in the TMA secondary decenters (≈ 1.5 mm) is comparable to the half-width of the DE bounds used in Chapter 4.

5.5 Observations after known moves of the secondary

Section 5.3 showed that the Hubble DE estimate is limited by the observation noise, amplified along the decenter–tilt degeneracy. There are two ways to raise its precision: collect more light, which reduces every error as $1/\sqrt{M}$ for M exposures, or make the observations themselves more informative. Phase diversity (Gonsalves, 1982) does the latter by recording a second image with a known added aberration, usually defocus. The alignment mechanism can provide the added aberration itself: the secondary is moved by known amounts relative to its unknown position, and the PSFs are recorded again at each position. This section asks how much such moves improve the Hubble estimate, first with the linearized analysis of Section 5.3 and then with full runs of the pipeline for two move sizes.

If the secondary is observed at $\boldsymbol{\theta}_{\text{true}} + \boldsymbol{\delta}_m$, $m = 0, \ldots, M-1$, with $\boldsymbol{\delta}_0 = \mathbf{0}$ and every $\boldsymbol{\delta}_m$ known, the DE objective (equation 2.10) becomes a sum over positions,

$$S(\boldsymbol{\theta}) = \sum_{m=0}^{M-1} \sum_{f \in \mathcal{F}} \sum_{k=1}^{N_z} \left[c_{f,k}(\boldsymbol{\theta} + \boldsymbol{\delta}_m) - c_{m,f,k,\text{obs}} \right]^2 , \tag{5.4}$$

and equation (5.3) becomes

$$\text{Cov}(\hat{\boldsymbol{\theta}}) = \Big(\sum_m \mathbf{J}_m^\top \mathbf{J}_m \Big)^{-1} \Big(\sum_m \mathbf{J}_m^\top \mathbf{N}_{\text{all}} \mathbf{J}_m \Big) \Big(\sum_m \mathbf{J}_m^\top \mathbf{J}_m \Big)^{-1}, \tag{5.5}$$

where $\mathbf{J}_m$ is the Jacobian of the coefficients of all fields at $\boldsymbol{\theta}_{\text{true}} + \boldsymbol{\delta}_m$ and $\mathbf{N}_{\text{all}}$ is as before. If every $\boldsymbol{\delta}_m$ is zero, the M exposures are repeats, all $\mathbf{J}_m$ are equal and every standard deviation falls by $\sqrt{M}$. The benefit of moving is therefore measured against the same number of

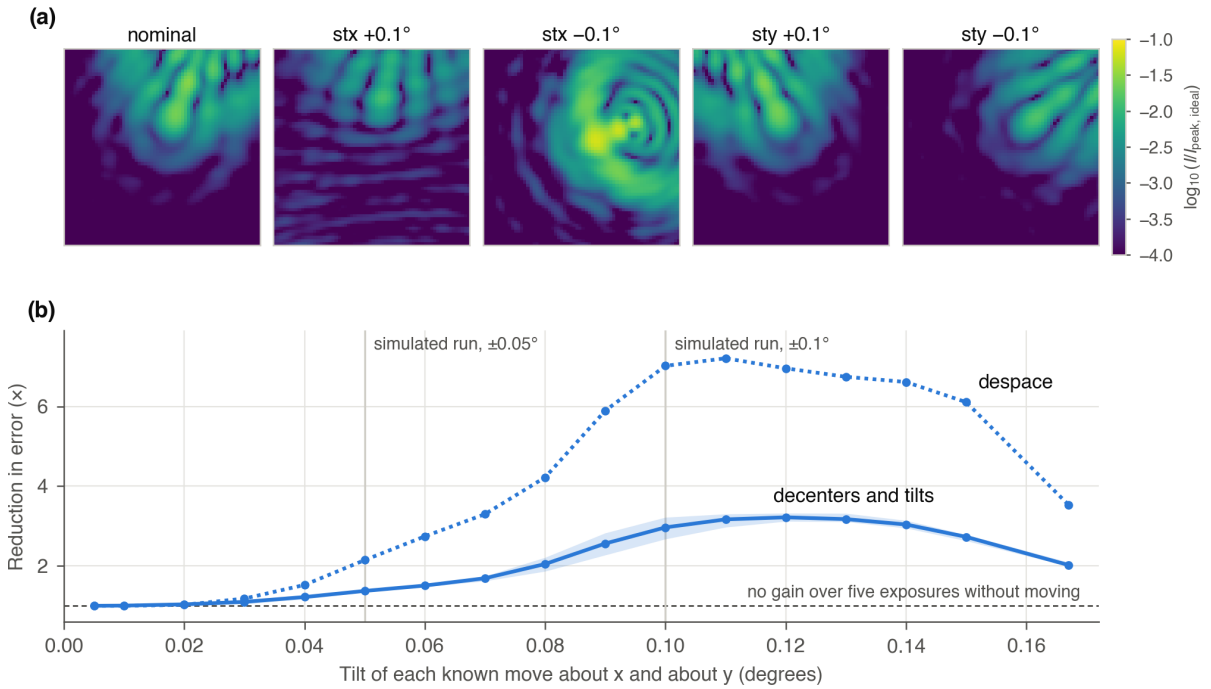


Figure 5.3: Known moves of the Hubble secondary. (a) On-axis PSF, on a logarithmic scale relative to the ideal peak, at the injected misalignment and after the four known tilts of ±0.1 degrees used in the first simulation. (b) Predicted reduction in the noise-limited error of the DE estimate relative to five exposures at the nominal position (equation 5.5), for five positions: the nominal one and tilts of $\pm\Delta$ about x and about y. The band spans the four decenter and tilt parameters and the solid line is their mean; the dotted line is the despace. The vertical lines mark the two simulated designs.

exposures at the nominal position: we define the gain as the standard deviation from M such exposures divided by the standard deviation with the moves.

Figure 5.3b shows the gain for designs of five positions: the nominal one and tilts of $\pm\Delta$ about x and about y. Moves of 0.01 degrees or less give no gain, and moves of 0.02 degrees at most 7%. The PSF responds almost linearly to such small changes, so a moved PSF carries nearly the same information as a repeated exposure, and the improvement is essentially the factor $\sqrt{5}$ from five times the exposure. The gain rises from about 0.03 degrees, peaks near 0.11–0.12 degrees at 3.1–3.3 on the decenters and tilts and 7.2 on the despace, and falls for larger moves, possibly because these push the PSF toward the edge of the 64 × 64-pixel cutout (Figure 5.3a). What matters is the amount of known coma a move adds rather than the actuator that makes it. In coma, one degree of secondary tilt is equivalent to about 10 mm of decenter, and decenter moves matched in coma give similar gains: 2.3–2.7 on the

decenters and tilts for ±1 mm, but none for ±100–200 μm. Moves in despace mainly improve the despace estimate (by 1.9 for ±100 μm, against at most 1.2 for the other parameters), and moves of ±400 μm degrade every parameter.

To test the prediction, we repeated the Hubble benchmark of Chapter 3 twice, keeping the injected misalignment: with the nominal position and tilts of ±0.1 degrees about x and about y, close to the predicted optimum, and with tilts of ±0.05 degrees, for which the predicted gain from the moves is only 1.3–2.2. Each of the five positions was observed at the five fields of Section 3.1 with independent noise ($\sigma = 0.01$), giving 25 PSFs and 375 coefficients per run; the two runs share the same noise draws. The likelihood (equation 2.9) and the DE objective (equation 5.4) sum over the positions. Each moved position uses the calibration covariance $\mathbf{\Sigma}_f$ of its field; this is an approximation, but a 0.1-degree move lies well within the distribution from which $\mathbf{\Sigma}_f$ was estimated. The noise realization differs from that of Chapter 3. Since an evaluation now needs 25 PSFs, the five positions were computed in parallel, which kept the time per evaluation close to that of the nominal run.

With tilts of ±0.1 degrees, Nautilus needed 75 400 likelihood evaluations ($N_{\text{eff}} \approx 1.6 \times 10^4$). Unlike the nominal run, whose *sdx* and *sdy* marginals are nearly flat (Figure 3.3), the posterior concentrates on a lattice of modes about 100 μm apart in *sdx* and *sdy*, the highest of which contains the truth. The lattice comes from the prior transform, which rounds the tilts to 0.01 degrees (Section 2.2.4): a tilt step of 0.01 degrees is compensated by about 100 μm of decenter, so each neighbouring tilt value produces its own mode. The weighted median of the final 100 posterior samples was (53.17, −114.35, 39.05, 0.13, −0.04), within 1 μm of the truth on both decenters, compared with 113 μm on *sdy* in the nominal run. Its on-axis WFE, 1.9×10^{-3} waves, is already close to the final result of the nominal run. Because the tilt posterior is narrower than the rounding step, nearly all of the weight of the final 100 samples lies on the true tilt values, and the 2.1%–97.9% DE bounds on *stx* and *sty* collapse to single values. SciPy holds a parameter with equal bounds fixed, so we widened each collapsed bound by one rounding step on each side, to (0.12, 0.14) and (−0.05, −0.03) degrees.

With tilts of ±0.05 degrees, Nautilus needed 60 900 evaluations ($N_{\text{eff}} \approx 1.4 \times 10^4$). The posterior shows the same lattice of modes, but the modes carry similar weight, and most of

the weight of each tilt is spread over three grid values, 0.11–0.13 and −0.05 to −0.03 degrees. No bound collapsed, and the DE bounds on the decenters stayed wide, 220 μm on *sdx* and 125 μm on *sdy* (against 327 and 378 μm in the nominal run). The weighted median of the final 100 samples, (51.05, −111.11, 39.15, 0.13, −0.04), was still within 3 μm of the truth on both decenters, with an on-axis WFE of 5.2×10^{-3} waves.

In both runs DE used deferred updating, as in the TMA run, so that each generation could be evaluated in parallel. In the ±0.1-degree run, its cost came within 1% of its final value by generation 47, and the run was stopped by hand after 137 generations at $S = 2.78 \times 10^{-4}$. The expected minimum of the five-position objective is $(2.84 \pm 0.23) \times 10^{-4}$, so the run reached the noise floor; L-BFGS-B polishing left the solution unchanged. Table 5.4 compares the residuals with the predicted standard deviations. For this run, every residual lies within 0.9 standard deviations of zero, and the squared Mahalanobis distance of the residual vector is 2.9, a typical value for five degrees of freedom. The predicted standard deviations are 6.0–7.2 times smaller than for the nominal position alone on the decenters and tilts and 16 times smaller on the despace. Of these factors, $\sqrt{5} \approx 2.2$ comes from the additional exposures and 2.7–3.2 (7.0 on the despace) from the moves. Image quality improves accordingly. Propagating the covariance to the on-axis OPD by Monte Carlo (300 draws) gives a median RMS WFE of 1.5×10^{-4} waves (16th–84th percentiles 0.8–2.4×10^{-4}), against 1.2×10^{-3} waves for the nominal position. The run reached 2.1×10^{-4} waves, compared with 1.35×10^{-3} waves in Chapter 3.

The ±0.05-degree run was given its full 1000 generations. Its cost came within 1% of its final value by generation 48 and within 0.1% by generation 70, ending at 2.75×10^{-4} against an expected $(2.83 \pm 0.23) \times 10^{-4}$; polishing moved the estimate by less than 10^{-7}. Its residuals lie between 1.0 and 1.7 standard deviations from zero, larger than in the ±0.1-degree run but consistent with the noise. They come in pairs along the two degenerate directions, *sdy* with *stx* ($z = -1.54$ and $+1.54$) and *sdx* with *sty* ($z = -1.70$ and -1.63), and the squared Mahalanobis distance of the residual vector, 5.4, is again typical for five degrees of freedom. The predicted standard deviations are 3.0–3.1 times smaller than for the nominal position alone on the decenters and tilts and 4.8 times smaller on the despace. Of these factors, $\sqrt{5}$ comes from the additional exposures and only 1.3–1.4 (2.2 on the despace) from the moves,

Table 5.4: Noise-limited precision of the Hubble DE estimate from the nominal position alone (Table 5.2) and from the nominal position with tilts of ±0.05 or ±0.1 degrees about x and about y (predicted s.d., equation 5.5), with the residuals $\boldsymbol{\theta}_{\text{true}} - \mathbf{c}$ of the two five-position runs; z is the ratio of each residual to the predicted standard deviation of its run. Translations are in μm and tilts in degrees.

	Nominal	**Tilts of ±0.05°**			**Tilts of ±0.1°**		
Parameter	**s.d.**	**s.d.**	**Residual**	z	**s.d.**	**Residual**	z
sdx	3.62	1.21	−2.05	−1.70	0.61	0.16	0.26
sdy	3.96	1.29	−1.99	−1.54	0.56	−0.006	−0.01
sdz	0.075	0.016	−0.015	−0.97	0.0048	−0.0043	−0.90
stx	3.8×10^{-4}	1.2×10^{-4}	1.9×10^{-4}	1.54	5.3×10^{-5}	9.9×10^{-6}	0.19
sty	3.6×10^{-4}	1.2×10^{-4}	-1.9×10^{-4}	−1.63	5.7×10^{-5}	2.2×10^{-5}	0.39

as Figure 5.3b predicts. The predicted median on-axis WFE is 2.9×10^{-4} waves (16th–84th percentiles 1.6–4.3×10^{-4}), and the run reached 1.95×10^{-4} waves.

The moves do not break the decenter–tilt degeneracy. The correlations between the decenters and the perpendicular tilts are $r = -0.989$ for *sdy–stx* and $r = +0.995$ for *sdx–sty* with tilts of ±0.1 degrees, and −0.997 and +0.996 with tilts of ±0.05 degrees, against −0.996 and +0.995 for the nominal position; the standard deviations of the four decenters and tilts shrink by similar factors. The gain comes instead from sensitivity. The known coma added by each move makes the coefficients respond more strongly to every parameter, much as the known defocus of phase diversity does. Separating a decenter from the compensating tilt would still require observables that distinguish the two motions (Section 6.5). One such observable is the position of the image on the detector, which the present pipeline discards because each PSF is centered on the chief ray (Section 2.1). Along the degenerate direction the image moves a long way: a tilt of 0.02 degrees combined with the 204 μm decenter that compensates its coma changes the coefficients about a ninth as much as the tilt alone, but moves the image by 2.5 mm, about 780 pixels. In practice this displacement would have to be measured against the telescope's pointing reference. The two runs also show how little the degenerate directions matter for the image. The decenter errors of the ±0.05-degree run, about 2 μm, are more than ten times those of the ±0.1-degree run, yet its on-axis WFE is no larger (1.95 against 2.1×10^{-4} waves), because an error along a degenerate direction barely changes the on-axis wavefront.

The gain also depends on knowing the moves. In the linearized model, an unknown error $\boldsymbol{\epsilon}_m$ in move m shifts the estimate by $\left(\sum_k \mathbf{J}_k^\top \mathbf{J}_k\right)^{-1} \mathbf{J}_m^\top \mathbf{J}_m \boldsymbol{\epsilon}_m$. For the ±0.1-degree design, such errors become as large as the noise term once they reach about 0.8 μm in decenter, 5×10^{-5} degrees in tilt or 5 nm in despace per move. Less accurately known moves would have to be estimated as additional parameters, which would give up part of the gain. The analysis assumes that the moved positions stay within the range where $\boldsymbol{\Sigma}_f$ and the forward model apply, and it treats additive noise only (Section 5.1). The two full runs share one noise realization and one injected misalignment, and the most useful move size depends on the misalignment, since a move must be comparable to the coma already present to change the PSF appreciably.

5.6 Summary

At the noise level of the benchmarks, 10^{-4} of the ideal peak, the noise barely changes the likelihood ($-2 \ln \mathcal{L}$ shifts by less than 0.03 at the truth), so the nested-sampling posteriors reflect the calibration covariance rather than the measurement. The noise instead sets the precision of the DE estimate. Both runs reached the noise floor of their objectives, and their residuals and wavefront errors match the linearized predictions: about 4 μm on the Hubble decenters, 0.08 μm on its despace and $\sim 10^{-3}$ waves of WFE; about 50 μm on the TMA secondary decenters, ~ 2 μm on its despaces and a few 10^{-3} waves of WFE. Parameter degeneracies amplify the noise, by a factor of ~ 10 for the Hubble decenters and more than 100 for the TMA secondary decenters, relative to the precision if the other parameters were known. Reducing these degeneracies, for example with observables that distinguish a decenter from a compensating tilt, is therefore a natural route to higher precision. Observations after known moves of the secondary offer another route (Section 5.5). Observing the Hubble secondary again after tilts of ±0.1 degrees about each axis leaves the degeneracies in place but makes the coefficients more sensitive to every parameter: beyond the factor $\sqrt{5}$ gained from the additional exposures, the predicted errors fall by a further 2.7–3.2 on the decenters and tilts and 7.0 on the despace. A full run of the pipeline with these moves recovered both decenters to within 0.2 μm and reached 2.1×10^{-4} waves of on-axis WFE. Tilts of ±0.05 degrees gave a further factor of only 1.3–1.4 (2.2 on the despace), and moves of 0.01 degrees

or less give none. The errors grow linearly with the noise up to 3×10^{-3} of the peak for Hubble and up to 3×10^{-4} for the TMA; beyond that, the TMA errors grow faster than linearly and its WFE exceeds the Maréchal criterion near 1.7×10^{-3} of the peak. These conclusions apply to additive Gaussian noise; signal-dependent photon noise, background and detector effects remain to be studied.

Chapter 6

Conclusions and future work

This thesis asked whether the positions of a telescope's mirrors can be inferred directly from images of point sources. We presented a Bayesian pipeline that does this for simulated multi-field observations of telescopes with known prescriptions. A ray-trace model in Optiland applies rigid-body misalignments to the mirrors and computes the PSF at several field points. Each PSF is reduced to a short vector of basis coefficients, and a multivariate Gaussian likelihood compares these coefficients with the observed ones. Nested sampling with Nautilus explores the posterior over the alignment parameters, and differential evolution, started from and bounded by that posterior, refines a point estimate. This chapter summarizes the results (Section 6.1), discusses what they show about the two stages of the pipeline (Section 6.2) and about the limits of its precision (Section 6.3), states the limitations of the study (Section 6.4) and outlines future work (Section 6.5).

6.1 Summary of results

Two benchmarks tested the pipeline (Table 6.1). In both, the alignment was inferred from noisy observations, with Gaussian noise of 10^{-4} of the ideal PSF peak added to every observed image.

The first benchmark was a Ritchey–Chrétien model of the Hubble Space Telescope with five secondary-mirror degrees of freedom, observed at five field points and described by 15 circular-Bessel coefficients per field (Chapter 3). Nested sampling constrained the despace and both tilts but left the two decenters nearly unconstrained across the $\pm 200\,\mu$m prior. Differential evolution, searching within bounds taken from the posterior, recovered the trans-

lations to within $4\,\mu$m and the tilts to within about 4×10^{-4} degrees. It reduced the on-axis RMS wavefront error from 2.26 waves to 1.3×10^{-3} waves.

The second benchmark was a three-mirror anastigmat with ten degrees of freedom on its primary and secondary mirrors, observed at ten field points and described by 11 Zernike coefficients per field (Chapter 4). Its posterior was broader. The primary decenter *pdx* and both primary tilts were nearly unconstrained, the secondary decenters were broad, and several pairs of parameters were strongly correlated. Differential evolution recovered five of the six translations to within $10.3\,\mu$m and the secondary decenter *sdx* to within $46\,\mu$m. It reduced the on-axis RMS wavefront error from 8.24 waves to 2.5×10^{-3} waves.

Chapter 5 showed that these residuals are the ones the noise predicts. Both differential-evolution runs reached the noise floor of their objective, and every parameter residual lies within 1.6 predicted standard deviations of zero. The final wavefront errors also lie within the range obtained by propagating the noise through the linearized forward model. The remaining errors therefore come from the noise in the observations, not from the optimizer.

Chapter 5 also tested a way to improve on this limit for Hubble: observing the secondary again after known moves (Section 5.5). With four tilts of ± 0.1 degrees, the predicted noise-limited errors were 6–7 times smaller than from the nominal position alone, and 16 times smaller on the despace; a factor of $\sqrt{5}$ comes from the four additional exposures and the rest from the moves. A full run of the pipeline recovered both decenters to within $0.2\,\mu$m and reduced the on-axis RMS wavefront error to 2.1×10^{-4} waves.

6.2 The roles of the two stages

The two stages of the pipeline do different jobs, and the benchmarks show that both are needed. Nested sampling is the global search. Starting from the full prior, it found the high-likelihood region in both benchmarks, and its starting point alone reduced the on-axis wavefront error by a factor of about 160 for Hubble and about 8 for the TMA. The posterior also maps the structure of the problem. Its marginals show which parameters the observations constrain, and its correlations reveal combinations of mirror motions that leave the PSFs nearly unchanged. An optimizer alone would return a single point without this

Table 6.1: Summary of the two benchmarks. WFE is the on-axis RMS wavefront error of the system with residual misalignment $\boldsymbol{\theta}_{\text{true}} - \mathbf{c}$ (Section 2.1). The noise-limited range is the 16th–84th percentile range predicted from the noise alone (Section 5.3).

	Hubble	**TMA**
Mirrors perturbed	secondary	primary, secondary
Degrees of freedom	5	10
Fields × modes	5×15	10×11
Likelihood evaluations (nested sampling)	44 000	93 200
Evaluations of S (DE, before polishing)	150 075	300 150
WFE, uncorrected (waves)	2.26	8.24
WFE, nested-sampling start $\mathbf{x}_0$ (waves)	0.014	0.99
WFE, DE solution (waves)	1.3×10^{-3}	2.5×10^{-3}
WFE, noise-limited range (waves)	$(0.7\text{–}1.7) \times 10^{-3}$	$(2.4\text{–}6.9) \times 10^{-3}$
Largest translation residual (μm)	3.8 (sdy)	46 (sdx)
Largest tilt residual (degree)	4.1×10^{-4} (stx)	3.7×10^{-4} (pty)

information. Nested sampling also computes the evidence, which we report (Table 2.3) but have not yet used.

The posterior does not, however, give the precision of the measurement. Its width is set by the calibration covariances $\boldsymbol{\Sigma}_f$, which describe how the coefficients vary across the prior rather than the measurement noise; at the noise level of the benchmarks, the noise shifts $-2 \ln \mathcal{L}$ by less than 0.03 at the truth (Section 5.2). Differential evolution supplies the precision. By minimizing the unweighted coefficient residuals, it reduced the wavefront error by a further factor of about 10 for Hubble and about 400 for the TMA and reached the noise floor in both runs. Over both stages, the wavefront error fell by factors of about 1700 and 3300.

Nested sampling also made this refinement tractable. The bounds taken from the innermost posterior shell narrowed the search in the well-constrained parameters by more than an order of magnitude, for example from $\pm 200\,\mu$m to 36–43 μm on the Hubble despace and from ± 0.4 degrees to ranges 0.02–0.03 degrees wide on the TMA secondary tilts, and contained the true alignment in both runs. L-BFGS-B polishing left both solutions unchanged, but this says little about convergence, because its finite-difference gradients are dominated by numerical noise in the objective (Section 2.3). That both objectives ended at their predicted noise floors (Section 5.3) is what shows that differential evolution reached the minimum.

6.3 What limits the precision

The precision of the recovered alignment is set by the noise and by the near-degeneracies of the problem together. Near-degenerate combinations of mirror motions appeared in both benchmarks, and they have a simple optical origin. A decenter of a mirror and a tilt about the perpendicular axis, of the same mirror or of another, both introduce coma that is nearly constant across the field, so one can largely cancel the other. This links *sdy* with *stx* and *sdx* with *sty* for Hubble, and *sdy* with *ptx* and *sdx* with *pty* and *sty* for the TMA. The two despaces of the TMA trade off against each other in setting focus. These compensating motions are the ones described by nodal aberration theory (Thompson, 2005; Schechter and Levinson, 2011). A compensating motion cancels the field-constant aberration but leaves a field-dependent residual, which is why the likelihood must combine several field points.

The linearized analysis of Chapter 5 quantified what these degeneracies cost. If all other parameters were known, the noise alone would determine the Hubble decenters to 0.36 and 0.32 μm and the TMA secondary decenter *sdx* to 0.30 μm. The correlations amplify these errors by a factor of about 10 for Hubble and about 170 for the TMA. The largest residual of the TMA run, 46 μm on *sdx*, is therefore noise projected onto a nearly degenerate direction rather than a failure of the optimizer.

Because the degenerate directions change the PSFs little, errors along them also change the image quality little: the TMA solution, with its 46 μm error in *sdx*, leaves only 2.5×10^{-3} waves of on-axis wavefront error. The two Hubble runs with known moves show the same effect: their decenter errors differ by more than a factor of ten, yet they leave the same on-axis wavefront error, about 2×10^{-4} waves (Section 5.5). For correcting the image quality at the observed fields, the degeneracies matter less than the parameter residuals suggest. They matter more when the mirror positions themselves are needed, for example to track mechanical drifts, or when the aberrations at field points that were not observed must also be controlled.

The noise sets the scale of these errors. At the noise level of the benchmarks, the parameter errors and the wavefront error grow in proportion to the noise up to 3×10^{-3} of the ideal peak for Hubble and up to 3×10^{-4} for the TMA (Section 5.4). Beyond that, the TMA

despace errors grow faster than linearly as the trade-off between the two despaces becomes non-linear, and the linear prediction places the median wavefront error at the Maréchal criterion of $\lambda/14$ near 1.7×10^{-3} of the peak. Within the assumptions of this study, the pipeline could therefore bring a TMA of this design to a median wavefront error within the Maréchal criterion from observations whose noise is below about 0.17% of the ideal PSF peak.

The precision can also be raised by making the observations more informative rather than less noisy. Moving the Hubble secondary by known tilts and observing again adds known coma to the PSFs, which makes the coefficients respond more strongly to every parameter, much as the known defocus of phase diversity does (Section 5.5). With tilts of ± 0.1 degrees, the predicted errors fell by a further factor of about 3 on the decenters and tilts, and 7 on the despace, beyond the $\sqrt{5}$ gained from the additional exposures. The moves did not, however, weaken the degeneracies: the decenter–tilt correlations stayed above 0.98 in magnitude. Their benefit depends on their size, since a move must be comparable to the coma already present to change the PSF appreciably; tilts of ± 0.05 degrees gave a further factor of only 1.3–1.4, and tilts of 0.01 degrees or less none. It also depends on knowing the moves, to about $0.8\,\mu$m in decenter, 5×10^{-5} degrees in tilt and 5 nm in despace per move for the ± 0.1-degree design.

6.4 Limitations

The benchmarks are idealized in several ways, and the results should be read with these in mind.

- **The forward model is exact.** The observations were generated with the same Optiland model used for inference, so the prescriptions, the aperture masks and the positions of the fixed optics and the detector are known exactly. A real telescope differs from its design through figure errors on the mirrors, uncertainties in their radii and conic constants, and misplaced fixed components. Any such mismatch would bias the inferred positions, and its effect has not been measured here.

- **The PSFs are simplified.** They are monochromatic, sampled at 4 and 8 pixels per $\lambda F_{\#}$, and free of detector effects such as pixel integration and jitter. They are also centered on the chief ray, so the position of the image on the detector is not used. The noise is additive, Gaussian and independent of the signal; photon noise, background and detector systematics are not modeled.

- **Each benchmark is a single case.** Each telescope was tested with one injected misalignment and one noise realization per field. Chapter 5 shows that the residuals are typical of the noise, but the success rate of the pipeline over many misalignments, in particular large ones near the edges of the prior, has not been measured.

- **The posterior widths are not measurement uncertainties.** Because the likelihood uses the calibration covariances $\mathbf{\Sigma}_f$, the credible intervals in Figures 3.3 and 4.3 describe how the coefficients vary across the prior; the uncertainty of the final estimate comes from the linearized analysis of Chapter 5, not from the sampler. The Hubble prior transform also rounds the tilts to 0.01 degrees, far coarser than the precision that differential evolution reached. In the runs with known moves of the secondary, this rounding split the posterior into separate modes about 100 μm apart in the decenters and, with tilts of ± 0.1 degrees, collapsed the differential-evolution bounds on the tilts onto single values, which had to be widened by hand (Section 5.5).

- **Only some mirrors move.** The Hubble model perturbs only the secondary, and the TMA only the primary and secondary. The tertiary, the fold mirror and the detector stay fixed, and rotations about the optical axis are not included.

- **The computation is expensive.** Counting one PSF per field for every likelihood evaluation and every evaluation of the DE objective, the Hubble benchmark computed about 10^6 PSFs and the TMA about 4×10^6, roughly three quarters of them during differential evolution.

6.5 Future work

6.5.1 Realistic observations

The next step toward applying the pipeline is to make the simulated observations realistic and then to replace them with measured PSFs. The forward model can be extended to broadband PSFs by summing monochromatic PSFs over the passband, and to real detectors by integrating over pixels at the plate scale of an instrument. The noise model should include photon noise, read noise and background. For bright stars, photon noise is close to Gaussian with a variance that changes across the image; because the basis projection is linear, the resulting coefficient covariance follows from equation (5.2) with σ^2 moved inside the sum over pixels. The effect of model mismatch can be measured by injecting errors that the inference model does not contain, such as mirror figure errors or small changes in the conic constants, and recording the bias they cause in the inferred positions. The most important of these errors could then be added to the model as nuisance parameters and marginalized over. Finally, the pipeline should be applied to laboratory testbeds with known, actuated misalignments and then to on-sky PSF campaigns relevant to Habitable Worlds Observatory class TMA designs (NASA Goddard Space Flight Center, 2023).

6.5.2 A likelihood based on the noise

At present, the posterior locates the solution and differential evolution measures it. A likelihood built from the noise covariance $\mathbf{N}$ would instead give posterior widths that are measurement uncertainties, with the correlations of Figure 5.1 built in. Such a likelihood is far more sharply peaked, so a practical scheme would keep the present nested-sampling stage as the global search and sample the noise-based likelihood locally, for example with Markov chain Monte Carlo started from the differential-evolution solution. At the benchmark noise level, the linear predictions of Chapter 5 held to within about 20%, which suggests that this posterior would be close to the Gaussian of equation (5.3). A likelihood that describes the measurement would also make the evidence useful: comparing models in which different subsets of the degrees of freedom are free could identify which mirrors moved.

6.5.3 Breaking the degeneracies

The linearized analysis of Chapter 5 can also guide the design of observations. The covariance $\mathbf{A}\,\mathbf{N}_{\mathrm{all}}\,\mathbf{A}^{\top}$ requires only the Jacobian of the coefficients at a nominal alignment, so it can be evaluated for any candidate set of field points, basis or noise level before data are taken. It could be used to choose field points that best separate a decenter from a compensating tilt; the field-dependent aberrations that distinguish them grow with field angle. Other observables could break the degeneracies more directly: the position of each image on the detector, which the present pipeline discards because its PSFs are centered on the chief ray, although it moves by millimeters along the Hubble decenter–tilt degeneracy (Section 5.5); defocused or phase-diverse images (Gonsalves, 1982); PSFs at several wavelengths; or more basis modes, which would retain information that the present 11- and 15-mode projections discard. Section 5.5 tested one form of diversity, PSFs recorded after known moves of the secondary, and chose the size of the moves with this calculation. The moves raised the sensitivity of the coefficients severalfold but left the decenter–tilt correlations unchanged, so diversity of this kind complements, rather than replaces, observables that separate a decenter from a compensating tilt. For the TMA, where the degeneracies amplify the noise by more than 100, the same calculation could identify which moves of which mirror would help most. Where they are available, actuator or metrology readings could also constrain individual mirrors through the prior. Each of these options can be assessed with the same linear calculation before it is added to the full pipeline.

6.5.4 More degrees of freedom

Aligning a real observatory involves more degrees of freedom than the benchmarks: the tertiary and fold mirrors, the detector position, rotations about the optical axis for optics that are not rotationally symmetric, low-order changes in the figure of each mirror and, for segmented primaries such as those of JWST and several HWO concepts, the piston and tip-tilt of every segment. Doubling the number of parameters from Hubble to the TMA already broadened the posterior, strengthened the degeneracies and roughly doubled the number of likelihood evaluations. Scaling to tens or hundreds of parameters will need the efficiency

gains described below and the degeneracy breaking described above. A staged approach, in which the largest motions are solved first and finer terms are added around that solution, would follow the sequence of JWST commissioning (Acton et al., 2022).

6.5.5 Computational cost

Several changes could reduce the cost of the pipeline. Neither differential-evolution run met its convergence tolerance, because the population approaches the noise floor only slowly, and both ran the full 2000 generations although the Hubble objective was within 4% of its final value by generation 30. In the linear approximation, a parameter error of one standard deviation, in any direction, raises the objective by about the noise variance of a single coefficient. A stopping rule that compares recent improvements of the objective with this variance would end a run once further generations could no longer change the estimate meaningfully.

Differential evolution could also be replaced, after nested sampling, by a local least-squares solver. The Monte Carlo study of Section 5.4 solved the same least-squares problem with twelve Gauss–Newton iterations started from the truth. In practice, each iteration would need a new Jacobian, or $2d+1$ forward-model evaluations by central differences, so a few tens of iterations cost hundreds of evaluations rather than the $1.5\text{–}3 \times 10^5$ used by differential evolution. A damped Gauss–Newton solver, as in merit-function regression (Kim et al., 2007), started from the nested-sampling point could therefore reach the noise floor at a small fraction of the cost. A test on the Hubble benchmark, with a new noise realization, supports this. Started from the nested-sampling point, L-BFGS-B with gradients from Optiland's PyTorch backend came within 0.1 noise standard deviations of the differential-evolution solution after 31 evaluations of S, against about 5900 for differential evolution. Seven further starts spread over the nested-sampling bounds converged to the same minimum, and so did L-BFGS-B with central-difference gradients, with about ten times as many evaluations. SciPy's default forward-difference gradients failed from every start, for the reason given in Section 2.3. All the starts lay within the nested-sampling bounds, so nested sampling is still needed to provide them.

The forward model itself could be made faster. Every likelihood evaluation is independent, so both stages parallelize; the TMA run already evaluated each DE generation in parallel. Ray tracers with GPU and automatic-differentiation backends, including Optiland's PyTorch backend, could trace many misalignments at once and provide exact gradients for local refinement and for gradient-based samplers such as Hamiltonian Monte Carlo. In Optiland 0.6.0, however, the gradients with respect to decenters and tilts are correct only after a small fix. The reference sphere for the wavefront is built from detached scalar values, so automatic differentiation misses its dependence on the misalignment and returns gradients of the wrong sign that are 130–1900 times too large. Keeping the sphere in the computation graph makes them agree with central differences to three or more digits. Finally, the calibration step already computes 10^4 forward-model evaluations per field. A neural-network emulator of the coefficients trained on such draws could replace the ray trace inside the sampler, with ray tracing kept for the final refinement and validation. Simulation-based inference could go further and learn the posterior as a function of the observed coefficients, so that each new observation could be analyzed without a new sampling run.

6.5.6 Toward closed-loop alignment

In operation, alignment is a sequence of measurements rather than a single one: a telescope drifts, for example as its temperature changes, and is corrected repeatedly. Each update could use the previous solution and its uncertainty as the prior, which would narrow the search and reduce its cost. Combining the inferred positions with a model of the mirror actuators would close the loop: observe, infer the mirror positions, command corrective moves and observe again. A laboratory testbed with actuated mirrors is the natural place to demonstrate this loop.

6.6 Closing remarks

Phase retrieval and wavefront sensing show that a telescope is misaligned; this work shows that, for simulated telescopes with known prescriptions, the misalignment itself can be inferred from multi-field PSFs. In both benchmarks the recovered alignment left a few thou-

sandths of a wave of on-axis wavefront error, and the remaining errors in the mirror positions are those set by the noise and amplified along the near-degenerate directions of the problem. The approach is computationally intensive, but it measures the state of the optical system directly, in principle from a single image of a wide-field instrument, as is typical of modern TMA observatories, rather than through a linearized model of how the wavefront responds to mirror motions. With realistic observations, faster forward models, known mirror moves that sharpen the measurement and observables chosen to break the degeneracies, it could enable deterministic alignment of future observatories such as HWO (NASA Goddard Space Flight Center, 2023) and Lazuli (Roy et al., 2026).

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